

I N D E X

UNIT 1 – COMPLEX FUNCTION AND CONFORMAL MAPPING.....	1
METHOD – 1: BASIC EXAMPLES.....	3
METHOD – 2: SQUARE ROOT OF COMPLEX NUMBER.....	7
METHOD – 3: N^{TH} ROOT OF COMPLEX NUMBER.....	8
METHOD – 4: TRIGONOMETRIC FUNCTION OF COMPLEX NUMBER	12
METHOD – 5: LOGARITHM OF COMPLEX NUMBER.....	12
METHOD – 6: DIFFERENTIABILITY OF COMPLEX FUNCTION.....	15
METHOD – 7: ANALITICITY OF COMPLEX NUMBER.....	18
METHOD – 8: TO FIND HARMONIC FUNCTION.....	21
METHOD – 9: FIXED POINT, CRITICAL POINT, ORDINARY POINT.....	23
METHOD – 10: ELEMENTARY TRANSFORMATION.....	24
METHOD – 11: BILINEAR TRANSFORMATION	26
UNIT-2 » COMPLEX INTEGRAL, SEQUENCE AND SERIES.....	29
METHOD – 1: LINE INTEGRAL	30
METHOD – 2: MAXIMUM MODULUS THEOREM	33
METHOD – 3: CAUCHY INTEGRAL THEOREM	34
METHOD – 4: CAUCHY INTEGRAL FORMULA.....	36
METHOD – 5: CONVERGENCE OF A SEQUENCE.....	39
METHOD – 6: CONVERGENCE OF SERIES.....	41
METHOD – 7: REDIUS OF CONVERGENCE.....	42
METHOD – 8: TAYLOR'S SERIES AND MACLAURIN'S SERIES	44
UNIT-3 » LAURENT'S SERIES AND RESIDUES	47
METHOD – 1: LAURENT'S SERIES.....	47
METHOD – 2: RESIDUES OF FUNCTION	50
METHOD – 3: CAUCHY'S RESIDUE THEOREM	52

I N D E X

METHOD – 4: CONTOUR INTEGRATION BY USING RESIDUE THEOREM.....	55
UNIT-4 » FIRST ORDER PARTIAL DIFFERENTIAL EQUATION.....	59
METHOD – 1: EXAMPLE ON FORMATION OF PARTIAL DIFFERENTIAL EQUATION	60
METHOD – 2: EXAMPLE ON LAGRANGE’S DIFFERENTIAL EQUATION	63
METHOD – 3: EXAMPLE ON NON-LINEAR PDE	64
METHOD – 4: EXAMPLE ON CHARPIT’S METHOD	66
UNIT-5 » HIGHER ORDER PARTIAL DIFFERENTIAL EQUATION	67
METHOD – 1: EXAMPLE ON SOLUTION OF HOMO. HIGHER ORDERED PDE.....	68
METHOD – 2: EXAMPLE ON SOLUTION OF NON-HOMO. HIGHER ORDERED PDE	70
METHOD – 3: EXAMPLE ON SEPARATION OF VARIABLES.....	72
METHOD – 4: EXAMPLE ON CLASSIFICATION OF 2 ND ORDER PDE.....	74
METHOD – 5: EXAMPLE ON WAVE, HEAT AND LAPLACE EQUATION.....	82
GTU PAPERS	***
SYLLABUS OF CVPDE – 3130005.....	***

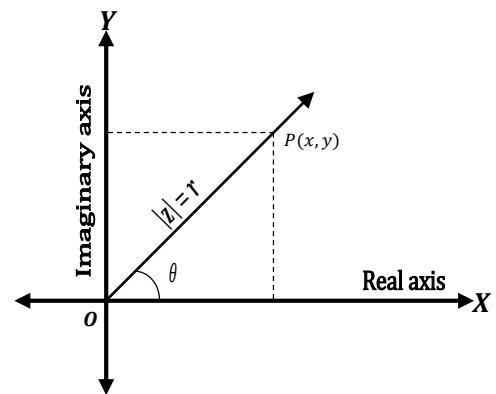
UNIT 1 – COMPLEX FUNCTION AND CONFORMAL MAPPING

❖ INTRODUCTION:

- ✓ A number $z = x + iy$ is called a complex number, where $x, y \in \mathbb{R}$ and $i = \sqrt{-1}$.
- ✓ x is called the real part of z and is denoted by $\text{Re}(z)$.
- ✓ y is called the imaginary part of z and is denoted by $\text{Im}(z)$.
- ✓ set $\mathbb{C} = \{x + iy / x, y \in \mathbb{R}, i = \sqrt{-1}\}$.
- ✓ Conjugate of a complex number $z = x + iy$ is denoted by $\bar{z}$ and is defined by $\bar{z} = x - iy$.
- ✓ Two complex number $x + iy$ and $x - iy$ are said to be complex conjugate of each other.

❖ GEOMETRICAL REPRESENTATION OF COMPLEX NUMBER:

- ✓ Let XOY to be a complex plane, where $\overrightarrow{OX}$ and $\overrightarrow{OY}$ are called **Real axis** and **Imaginary axis** respectively.
- ✓ The ordered pair $P(x, y)$ represents the complex number $z = x + iy$. The xy – plane is now known as **Argand plane** or **Complex plane** or **Gaussian Plane**.



- ✓ $\overline{OP}$ represents the distance between P and O , it is called **modulus** of z and denoted by $|z|$.

$$\text{i. e. } |z| = r = \sqrt{x^2 + y^2} = \sqrt{z \cdot \bar{z}}$$

- ✓ Let $\overline{OP}$ makes an angle θ with positive real axis, it is called **argument** of z .

$$\text{i. e. } \theta = \tan^{-1} \left(\frac{y}{x} \right)$$

❖ NOTES:

- (1). If $-\pi < \theta \leq \pi$, then argument of z is called “**PRINCIPAL ARGUMENT**” of z . It is denoted by **Arg(z)**.
- (2). **Arg(z)** is a Single-Valued Function.
- (3). The “**GENERAL ARGUMENT**” of argument of z is denoted by **arg(z)**.

$$\arg(z) = \tan^{-1} \left(\frac{y}{x} \right) + 2k\pi ; k = 0, \pm 1, \pm 2, \dots$$

(4). $\arg(z)$ is a Multi-Valued Function.

(5). For $z = 0 = 0 + i0$, argument is not defined.

❖ **RULES TO DETERMINE ARGUMENT OF A NON-ZERO COMPLEX NUMBER:**

- ✓ If $x > 0$ & $y > 0$, $\theta = \tan^{-1} \left(\frac{y}{x} \right)$
- ✓ If $x > 0$ & $y < 0$, $\theta = -\tan^{-1} \left| \frac{y}{x} \right|$
- ✓ If $x < 0$ & $y > 0$, $\theta = \pi - \tan^{-1} \left| \frac{y}{x} \right|$
- ✓ If $x < 0$ & $y < 0$, $\theta = -\pi + \tan^{-1} \left| \frac{y}{x} \right|$

❖ **ARITHMETIC OPERATIONS OF COMPLEX NUMBERS:**

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ be two complex numbers then

$$\mapsto \text{Addition: } z_1 + z_2 = (x_1 + iy_1) + (x_2 + iy_2) = (x_1 + x_2) + i(y_1 + y_2)$$

$$\mapsto \text{Subtraction: } z_1 - z_2 = (x_1 + iy_1) - (x_2 + iy_2) = (x_1 - x_2) + i(y_1 - y_2)$$

$$\mapsto \text{Multiplication: } z_1 \cdot z_2 = (x_1 + iy_1) \cdot (x_2 + iy_2) = (x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1)$$

$$\mapsto \text{Division: } \frac{z_1}{z_2} = \frac{x_1 + iy_1}{x_2 + iy_2} \times \frac{x_2 - iy_2}{x_2 - iy_2} = \frac{(x_1x_2 + y_1y_2) + i(x_2y_1 - x_1y_2)}{x_2^2 + y_2^2}$$

❖ **PROPERTIES:**

✓ Let z_1 and z_2 be two complex numbers then

$$\overline{(\overline{z_1})} = z_1$$

$$\overline{\left(\frac{z_1}{z_2} \right)} = \frac{\overline{z_1}}{\overline{z_2}} ; z_2 \neq 0$$

$$|z_1| = |\overline{z_1}|$$

$$\frac{z_1 + \overline{z_1}}{2} = \operatorname{Re}(z_1)$$

$$\overline{z_1 \pm z_2} = \overline{z_1} \pm \overline{z_2}$$

$$\frac{z_1 - \overline{z_1}}{2i} = \operatorname{Im}(z_1)$$

$$\overline{z_1 \cdot z_2} = \overline{z_1} \cdot \overline{z_2}$$

$$z \cdot \overline{z} = x^2 + y^2 = |z|^2$$

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

$$|z_1 - z_2| \geq ||z_1| - |z_2||$$

$$|z_1 \cdot z_2| = |z_1| \cdot |z_2|$$

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$$

❖ **EULER FORMULA**

✓ $e^{i\theta} = \cos \theta + i \sin \theta$

✓ $e^{-i\theta} = \cos \theta - i \sin \theta$

❖ **POLAR REPRESENTATION OF A COMPLEX NUMBER:**

✓ Let $z = x + iy$ be a complex number. Let $x = r \cos \theta$ and $y = r \sin \theta$; $\theta \in (-\pi, \pi]$.

Now, $z = x + iy = r \cos \theta + i r \sin \theta = r(\cos \theta + i \sin \theta) = \mathbf{r \operatorname{cis} \theta}$

Thus, $z = r(\cos \theta + i \sin \theta)$ is called Polar representation of a complex number.

✓ By Euler Formula, $e^{i\theta} = \cos \theta + i \sin \theta$

Then, $z = r e^{i\theta}$ is called “Exponential representation”.

METHOD – 1: BASIC EXAMPLES

C	1	Find the Real & Imaginary part of $f(z) = z^2 + 3z$. Answer: $\operatorname{Re}(f(z)) = x^2 + 3x - y^2$, $\operatorname{Im}(f(z)) = 2xy + 3y$	S 2020 (4)
C	2	Find the Real & Imaginary part of $f(z) = \frac{3i}{2 + 3i}$. Answer: $\operatorname{Re}(f(z)) = \frac{9}{13}$, $\operatorname{Im}(f(z)) = \frac{6}{13}$	W 2019 (3)
H	3	Find the value of $\operatorname{Re}(f(z))$ and $\operatorname{Im}(f(z))$ at the indicated point where $f(z) = \frac{1}{1 - z}$ at $7 + 2i$. Answer: $\operatorname{Re}(f(z)) = -\frac{3}{20}$, $\operatorname{Im}(f(z)) = \frac{1}{20}$	
H	4	Find the real part of $f(z) = \frac{1}{z - 2i}$ at point $1 + i$. Answer: $\operatorname{Re}(f(z)) = \frac{1}{2}$	
C	5	Separate real and imaginary parts of $f(z) = e^z$. Answer: $\operatorname{Re}(f(z)) = e^x \cos y$, $\operatorname{Im}(f(z)) = e^x \sin y$	
H	6	Separate real and imaginary parts of $\sinh z$. Answer: $\operatorname{Re}(\sinh z) = \sinh x \cos y$, $\operatorname{Im}(\sinh z) = \cosh x \sin y$	

C	7	Determine the modulus of following complex number. $z = 3 + 4i$; $z = \frac{1 - 2i}{i - 1}$. Answer: $5, \sqrt{\frac{5}{2}}$	
H	8	Determine the modulus of following complex number. $z = (4 + 2i)(-3 + \sqrt{2}i)$; $z = \frac{1 - 7i}{(2 + i)^2}$. Answer: $2\sqrt{55}, \sqrt{2}$	
C	9	Find the Principal Value of argument (Principal Argument). $z = -\sqrt{3} + i$; $z = -\sqrt{3} - i$; $z = \sqrt{3} - i$ Answer: $\frac{5\pi}{6}, -\frac{5\pi}{6}, -\frac{\pi}{6}$	
H	10	Find the Principal Value of argument (Principal Argument). $z = \frac{2 + 6\sqrt{3}i}{5 + \sqrt{3}i}$; $z = \frac{1 + 2i}{1 - 3i}$; $z = -\frac{2}{1 + \sqrt{3}i}$ Answer: $\frac{\pi}{3}, \frac{3\pi}{4}, \frac{2\pi}{3}$	
T	11	Find z if $\arg(z + 1) = \frac{\pi}{6}$ & $\arg(z - 1) = \frac{2\pi}{3}$. Answer: $\frac{1}{2} + \frac{\sqrt{3}}{2}i$	
T	12	Is $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$? Justify. Answer: No	
C	13	Express $\sqrt{3} - i$ into polar form. Answer: $2 \left(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right)$	
H	14	Express the following into polar form. $z = \left(\frac{2 + i}{3 - i} \right)^2$; $z = \frac{1 + i}{1 - i}$; $z = 3 + 4i$. Answer: $\frac{1}{2} \operatorname{cis} \left(\frac{\pi}{2} \right)$; $\operatorname{cis} \left(\frac{\pi}{2} \right)$; $5 \operatorname{cis} \left(\tan^{-1} \frac{4}{3} \right)$	

T	15	Prove that $ e^{-2z} < 1$ if and only if $\operatorname{Re}(z) > 0$.	
---	----	--	--

❖ **BASIC DEFINITIONS:**✓ **Distance:**

Let $z = a + ib$ and $w = c + id$ be complex numbers. Distance between z & w is defined as below.

$$\text{i.e. } |z - w| = \sqrt{(a - c)^2 + (b - d)^2}$$

✓ **Circle:**

If z' is a complex number and r is a positive number, then equation of circle is

$|z - z'| = r$. Where, z' is centre and r is radius of circle.

It gives the set of all those z whose distance from z' is r . [Points on the boundary]
[See Fig A].

✓ **Open Circular Disk:**

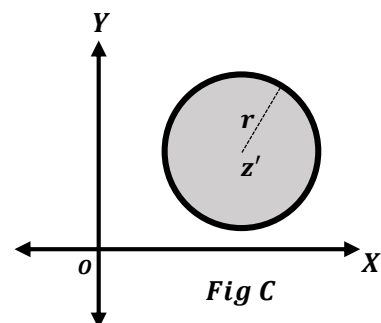
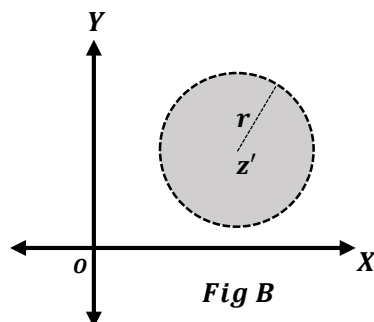
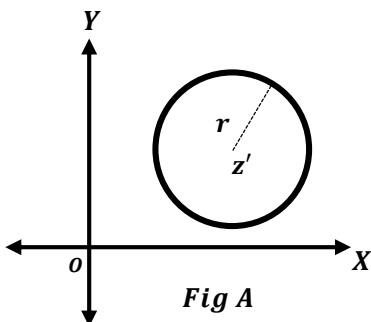
The equation $|z - z'| < r$ means set of all points inside the disk of radius r about z' .

Here, "OPEN" means that points on the boundary of circle are not in the set. [See Fig B]

✓ **Closed Circular Disk:**

The equation $|z - z'| \leq r$ means set of all points on the boundary and inside the disk of radius r about z' . It is union of circle and open circular disk.

Here, "CLOSED" means that points on the boundary of circle are in the set. [See Fig C]

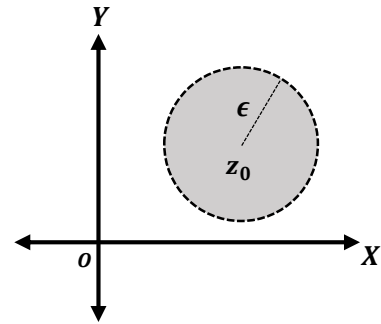


✓ **Neighborhood:**

The neighborhood of a point z_0 is set of points inside the circle centered at z_0 and radius ϵ .

$$\text{i. e. } |z - z_0| < \epsilon$$

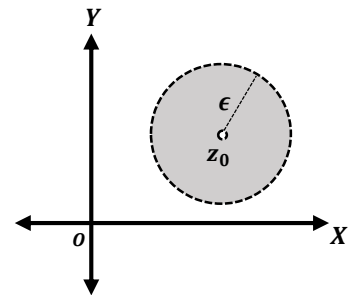
Neighborhood is nothing but a open circular disk with center z_0 and radius ϵ .

✓ **Deleted Neighborhood:**

The deleted neighborhood of a point z_0 is set of points inside the circle centered at z_0 and radius ϵ except the center z_0 .

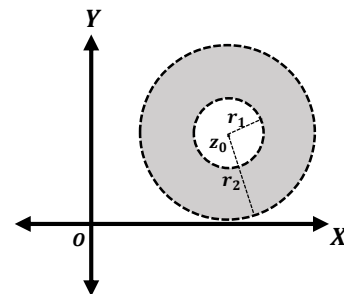
$$\text{i. e. } 0 < |z - z_0| < \epsilon$$

A deleted neighborhood is also known as “Punctured Disk”.

✓ **Annulus or Annular Region:**

The region between two concentric circle of radii r_1 & r_2 can be represented as

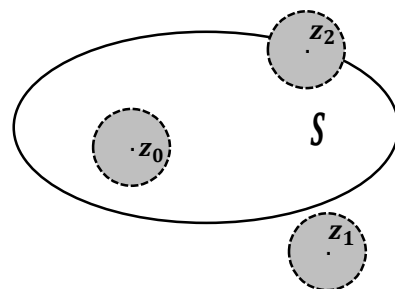
$$r_1 < |z - z_0| < r_2$$

✓ **Interior , Exterior and Boundary Points:**

A point z_0 is said to be interior point of a set S whenever there is some neighborhood of z_0 that contains only points of S .

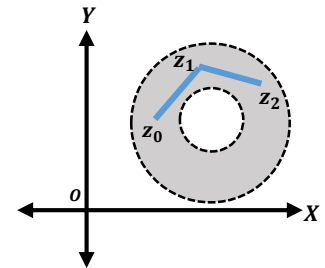
A point z_1 is said to be exterior point of a set S whenever there is no neighborhood of z_1 that contains only points of S .

A point z_2 is said to be boundary point of a set S whenever neighborhood of z_2 contains both interior and exterior points as well.



- ✓ **Open Set:** A set is open if it contains none of the boundary points.
- ✓ **Closed Set:** A set is said to be closed set if it contains all of the boundary points.
- ✓ **Connected Set:**

An open set S is connected if each pair of points z_0 and z_2 in it can be joined by a polygonal line, consisting of finite no. of line segments joined end to end that lies entirely in S .



- ✓ **Domain and Region:**

A set S is said to be domain if set S is open and connected. Note that any neighborhood is a Domain. A domain together with some, none or all of its boundary points is called region.

- ✓ **Bounded Region:**

A region is said to be bounded, if it can be enclosed in a circle of finite radius.

- ✓ **Compact region:**

A region is said to be compact if it is closed and bounded.

❖ FORMULA TO FIND SQUARE ROOT OF COMPLEX NUMBER:

- ✓ Let $z = x + iy$ be a complex number. Formula for finding square root of z is as below,

$$\sqrt{z} = \sqrt{x + iy} = \pm \left[\sqrt{\frac{|z| + x}{2}} + i(\text{sign of } y) \sqrt{\frac{|z| - x}{2}} \right]$$

METHOD – 2: SQUARE ROOT OF COMPLEX NUMBER

C	1	Find $\sqrt{-8 + 6i}$. Answer: $\pm (1 + 3i)$	
H	2	Find the roots of the equation $z^2 + 2iz + 2 - 4i = 0$. Answer: $z = 1 + i, -1 - 3i$	

C	3	Solve the Equation of $z^2 - (5 + i)z + 8 + i = 0$. Answer: $z = 3 + 2i, 2 - i$	S 2020 (4)
T	4	Find the roots of the equation $z^2 + (2i - 3)z + 5 - i = 0$. Answer: $z = 2 - 3i, 1 + i$	

❖ **DE-MOIVRE'S THEOREM:**

- ✓ Statement: $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$; $n \in \mathbb{Q}$ [i. e. $(e^{i\theta})^n = e^{in\theta}$]

❖ **PROCEDURE TO FINDING OUT NTH ROOT OF A COMPLEX NUMBER:**

- ✓ Convert into $z = r(\cos \theta + i \sin \theta)$; $r > 0$
- ✓ For, $n \in \mathbb{N}$

$$\begin{aligned} z^{\frac{1}{n}} &= r^{\frac{1}{n}} [\cos(\theta + 2k\pi) + i \sin(\theta + 2k\pi)]^{\frac{1}{n}} \\ &= r^{\frac{1}{n}} \left[\cos\left(\frac{\theta + 2k\pi}{n}\right) + i \sin\left(\frac{\theta + 2k\pi}{n}\right) \right] \\ &= r^{\frac{1}{n}} e^{i\left(\frac{\theta + 2k\pi}{n}\right)} ; k = 0, 1, 2, \dots, n - 1 \end{aligned}$$

Where, $r^{\frac{1}{n}}$ is positive nth root of r.

- ✓ By putting $k = 0, 1, 2, \dots, n - 1$, we have distinct roots of $z^{\frac{1}{n}}$.
- ✓ For $k = n, n + 1, n + 2, \dots$, we have repeated roots of $z^{\frac{1}{n}}$.

METHOD - 3: nth ROOT OF COMPLEX NUMBER

C	1	State De Moivre's formula. Find & plot 5 th roots of unity in complex plane. Answer: $z = e^{i\left(\frac{2k\pi}{5}\right)} ; k = 0, 1, 2, 3, 4$	
H	2	State De Moivre's formula. Find & plot 6 th roots of unity in complex plane. Answer: $z = e^{i\left(\frac{k\pi}{3}\right)} ; k = 0, 1, 2, 3, 4, 5$	

H	3	Find the fourth root of -1 . Answer: $z = e^{i\left(\frac{2k+1}{4}\right)\pi}; k = 0, 1, 2, 3$	
H	4	Find and plot the square root of $4i$. Answer: $z = \pm\sqrt{2}(1 + i)$	
C	5	Find the fourth root of -1 . Answer: $z = e^{i\left(\frac{2k+1}{4}\right)\pi}; k = 0, 1, 2, 3$	S 2020 (4)
C	6	State De Moivre's formula. Find and plot all root of $\sqrt[3]{8i}$. Answer: $z = 2 e^{i\left(\frac{4k+1}{6}\right)\pi}; k = 0, 1, 2$	
H	7	Find and plot all the roots of $(1 + i)^{\frac{1}{3}}$. Answer: $z = 2^{\frac{1}{6}} e^{i\left(\frac{\pi}{12} + \frac{2k\pi}{3}\right)}; k = 0, 1, 2$	
C	8	Show that if c is any n^{th} root of Unity other than Unity itself, then $1 + c + c^2 + \dots + c^{n-1} = 0$. OR Prove that the n roots of unity are in Geometric Progression. Also show that their sum is zero.	
T	9	Find the roots common to equation $z^4 + 1 = 0$ and $z^6 - i = 0$. Answer: $z = \frac{1-i}{\sqrt{2}}, \frac{-1+i}{\sqrt{2}}$	
H	10	Find the principal value of $\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^{\frac{3}{4}}$. Answer: $z = r \cdot e^{i\theta} = e^{i\frac{\pi}{4}} = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$	
C	11	Find real and imaginary part of $(-1 - i)^7 + (-1 + i)^7$. Answer: $\text{Re}(z) = -16; \text{Im}(z) = 0$	
H	12	State De Moivre's formula and hence evaluate $(1 + i\sqrt{3})^{100} + (1 - i\sqrt{3})^{100}$. Answer: $\text{Re}(z) = -2^{100}; \text{Im}(z) = 0$	W 2019 (4)

❖ **TRIGONOMETRIC (CIRCULAR) FUNCTIONS OF A COMPLEX NUMBER:**

✓ By Euler's Formula, $e^{iz} = \cos z + i \sin z \Rightarrow e^{-iz} = \cos z - i \sin z$

$$e^{iz} + e^{-iz} = 2 \cos z \Rightarrow \cos z = \frac{e^{iz} + e^{-iz}}{2}$$

$$e^{iz} - e^{-iz} = 2i \sin z \Rightarrow \sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

Hyperbolic Function Of a Complex Number	Relation between Circular and Hyperbolic Functions	
$\cosh z = \frac{e^z + e^{-z}}{2}$	$\sin ix = i \sinh x$	$\sinh ix = i \sin x$
$\sinh z = \frac{e^z - e^{-z}}{2}$	$\cos ix = \cosh x$	$\cosh ix = \cos x$
$\tanh z = \frac{e^z - e^{-z}}{e^z + e^{-z}}$	$\tan ix = i \tanh x$	$\tanh ix = i \tan x$
Hyperbolic Identities	Inverse Hyperbolic Functions	
$\cosh^2 x - \sinh^2 x = 1$	$\sinh^{-1} z = \log(z + \sqrt{z^2 + 1})$	
$\operatorname{sech}^2 x + \tanh^2 x = 1$	$\cosh^{-1} z = \log(z + \sqrt{z^2 - 1})$	
$\coth^2 x - \operatorname{cosech}^2 x = 1$	$\tanh^{-1} z = \frac{1}{2} \log\left(\frac{1+z}{1-z}\right)$	

✓ **Result:**

Show that $\sinh^{-1} z = \log(z + \sqrt{z^2 + 1})$, $\cosh^{-1} z = \log(z + \sqrt{z^2 - 1})$ and

$$\tanh^{-1} z = \frac{1}{2} \log\left(\frac{1+z}{1-z}\right)$$

Proof:

$$\text{Let, } w = \sinh^{-1} z \Rightarrow z = \sinh w = \frac{e^w - e^{-w}}{2}$$

$$\Rightarrow z = \frac{e^{2w} - 1}{2e^w}$$

$$\Rightarrow e^{2w} - 2ze^w - 1 = 0$$

$$\Rightarrow e^w = \frac{2z \pm \sqrt{4z^2 + 4}}{2} = z + \sqrt{z^2 + 1}$$

$$\Rightarrow w = \log(z + \sqrt{z^2 + 1})$$

$$\Rightarrow \sinh^{-1} z = \log(z + \sqrt{z^2 + 1}) \dots (A)$$

$$\text{Let, } w = \cosh^{-1} z \Rightarrow z = \cosh w = \frac{e^w + e^{-w}}{2}$$

$$\Rightarrow z = \frac{e^{2w} + 1}{2e^w}$$

$$\Rightarrow e^{2w} - 2ze^w + 1 = 0$$

$$\Rightarrow e^w = \frac{2z \pm \sqrt{4z^2 - 4}}{2} = z + \sqrt{z^2 - 1}$$

$$\Rightarrow w = \log(z + \sqrt{z^2 - 1})$$

$$\Rightarrow \cosh^{-1} z = \log(z + \sqrt{z^2 - 1}) \dots (B)$$

$$\text{Let } w = \tanh^{-1} z \Rightarrow z = \tanh w = \frac{\sinh w}{\cosh w} = \frac{e^w - e^{-w}}{e^w + e^{-w}}$$

$$\Rightarrow z = \frac{e^w - e^{-w}}{e^w + e^{-w}}$$

Taking componendo and dividendo, we get

$$\Rightarrow \frac{1+z}{1-z} = \frac{(e^w + e^{-w}) + (e^w - e^{-w})}{(e^w + e^{-w}) - (e^w - e^{-w})} = \frac{2e^w}{2e^{-w}} = e^{2w}$$

$$\Rightarrow 2w = \log\left(\frac{1+z}{1-z}\right)$$

$$\Rightarrow w = \frac{1}{2} \log\left(\frac{1+z}{1-z}\right)$$

$$\Rightarrow \tanh^{-1} z = \frac{1}{2} \log\left(\frac{1+z}{1-z}\right) \dots (C)$$

Eqn. (A), (B) & (C) are required equations.

METHOD – 4: TRIGONOMETRIC FUNCTION OF COMPLEX NUMBER

C	1	Prove that $\sin^{-1} z = -i \ln \left(iz + \sqrt{1 - z^2} \right)$	
T	2	Prove that $\cos^{-1} z = -i \ln \left(z + \sqrt{z^2 - 1} \right)$	
H	3	Prove that $\tan^{-1} z = \frac{i}{2} \log \frac{i + z}{i - z}$.	
T	4	Prove that $\operatorname{sech}^{-1} x = \log \left[\frac{1 + \sqrt{1 - x^2}}{x} \right]$.	
C	5	Show that $\cos(i\bar{z}) = \overline{\cos(iz)}$ for all z .	
H	6	Show that $\overline{\sin(iz)} = \sin(i\bar{z})$ if and only if $z = n\pi i$ ($n \in \mathbb{Z}$).	

❖ LOGARITHM OF A COMPLEX NUMBER:

✓ Polar representation of complex number, $z = re^{i\theta}$

$$\Rightarrow z = re^{i\theta}$$

$\Rightarrow \log z = \ln r + i\theta$ is called “GENERAL VALUE OF LOGARITHM”.

Where, r = Modulus of Z and θ = Argument of Z .

✓ If $\theta \in (-\pi, \pi]$, then it is called “PRINCIPAL VALUE OF LOGARITHM”.

❖ NOTE:

In Complex analysis,

(1). **Log** is used for Complex Single-Valued Function.

(2). **log** is used for Complex Multi-Valued Function.

(3). **ln** is used for Real Valued Function.

METHOD – 5: LOGARITHM OF COMPLEX NUMBER

C	1	Define: $\log(x + iy)$. Determine $\log(1 - i)$. Answer: $\ln \sqrt{2} + i \left(2k\pi - \frac{\pi}{4} \right)$	
---	---	--	--

H	2	Determine $\text{Log}(1 + i)$. Answer: $\ln \sqrt{2} + i \left(\frac{\pi}{4}\right)$	
T	3	Find the principal value of $z = \left[\frac{e}{2}(-1 - i\sqrt{3})\right]^{3\pi i}$. Answer: $3\pi i \left(1 - \frac{2\pi i}{3}\right)$	
C	4	Find all roots of the equation $\log z = \frac{i\pi}{2}$. Answer: $z = i$	S 2020 (3)
H	5	Define $\log z$, prove that $i^i = e^{-(4n+1)\frac{\pi}{2}}$.	W 2019 (4)
C	6	Find the value of $(-i)^i$. Answer: $e^{-(4n-1)\frac{\pi}{2}}$	

❖ **FUNCTION OF A COMPLEX VARIABLE:**

- ✓ If corresponding to each value of a complex variable $z = x + iy$ in a given region R , there correspond one or more values of another complex variable $w = u + iv$ then, w is called a function of the complex variable z and is denoted by

$$w = f(z) = u + iv$$

- ✓ Where, u and v are the real and imaginary part of w respectively and u and v are function of real variables x and y .

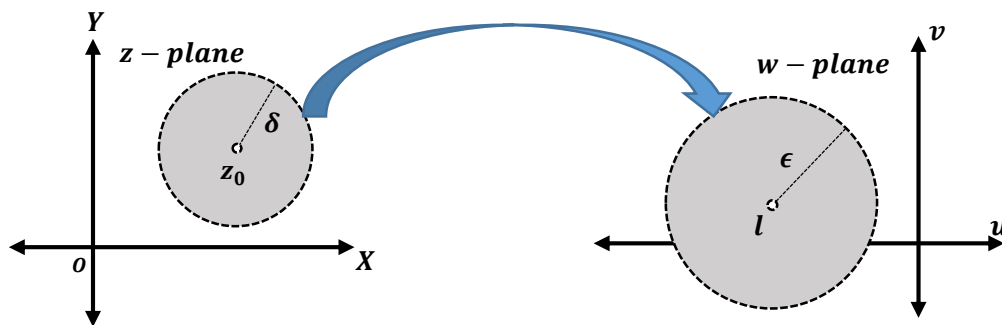
$$\text{i.e. } w = f(z) = u(x, y) + i v(x, y)$$

❖ **LIMIT OF COMPLEX FUNCTION:**

- ✓ A function $f(z)$ is said to have a limit l , if for each +ve number ϵ , there is +ve number δ such that

$$\text{i.e. } |f(z) - l| < \epsilon \text{ whenever } 0 < |z - z_0| < \delta$$

$$\text{Symbolically, } \lim_{z \rightarrow z_0} f(z) = l$$



❖ CONTINUITY OF COMPLEX FUNCTION:

- ✓ A complex valued function $f(z)$ is said to be continuous at a point $z = z_0$ if

$$(1). f(z_0) \text{ exists} \qquad (2). \lim_{z \rightarrow z_0} f(z) \text{ exists} \qquad (3). \lim_{z \rightarrow z_0} f(z) = f(z_0)$$

- ✓ Remark:

$f(z) = u(x, y) + i v(x, y)$ is continuous iff $u(x, y)$ and $v(x, y)$ are continuous.

If any one of these three conditions of continuity is not satisfied then $f(z)$ is discontinuous at $z = z_0$.

❖ DIFFERENTIABILITY OF COMPLEX FUNCTION:

- ✓ Let $w = f(z)$ be a continuous function and z_0 be a fixed point then $f(z)$ is said to be differentiable at z_0 if $\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$ exists, then the derivative of $f(z)$ at z_0 is denoted by $f'(z_0)$ and is defined as

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

- ✓ Another form:

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z + h) - f(z)}{h}$$

- ✓ Remark:

- (1). The rules of differentiation are same as in calculus of real variables.
- (2). If function is differentiable, then it is continuous.

METHOD – 6: DIFFERENTIABILITY OF COMPLEX FUNCTION

C	1	Prove $\lim_{z \rightarrow 1} \frac{iz}{3} = \frac{i}{3}$ by definition.	
H	2	Evaluate $\lim_{z \rightarrow i} \frac{z-i}{z^2+1}$. Answer: $-\frac{i}{2}$	
C	3	Show that the limit of the function does not exist $f(z) = \begin{cases} \frac{\text{Im}(z)}{ z } & ; z \neq 0 \\ 0 & ; z = 0 \end{cases}.$	
H	4	Discuss the continuity of $f(z) = \begin{cases} \frac{\bar{z}}{z} & ; z \neq 0 \\ 0 & ; z = 0 \end{cases}$ at origin. Answer: Limit does not exist , Discontinuous	
C	5	Discuss the continuity of $f(z) = \begin{cases} \frac{\bar{z}^2}{z} & ; z \neq 0 \\ 0 & ; z = 0 \end{cases}$ at $z = 0$. Answer: Limit exists , Continuous	
H	6	Discuss continuity of $f(z) = \begin{cases} \frac{\text{Re}(z^2)}{ z ^2} & ; z \neq 0 \\ 0 & ; z = 0 \end{cases}$ at $z = 0$. Answer: Limit does not exist , Discontinuous	
H	7	Find out and give reason whether $f(z)$ is continuous at $z = 0$, if $f(z) = \begin{cases} \frac{\text{Re}(z^2)}{ z } & ; z \neq 0 \\ 0 & ; z = 0 \end{cases}.$ Answer: Limit exists , Continuous	

C	8	Find the derivative of $\frac{z-i}{z+i}$ at i. Answer: $\frac{1}{2i}$	
H	9	Discuss the differentiability of $f(z) = x^2 + iy^2$. Answer: differentiable except the line $y = x$	
C	10	Show that $f(z) = z \operatorname{Im}(z)$ is differentiable only at $z = 0$ and $f'(0) = 0$.	
C	11	Show that $f(z) = z ^2$ is continuous at each point in the plane, but not differentiable.	
T	12	Show that $f(z) = \bar{z}$ is nowhere differentiable.	

❖ ANALYTIC FUNCTION:

✓ A function $f(z)$ is said to be analytic at point $z_0 = x_0 + iy_0$ if the function is differentiable at point z_0 as well as it is differentiable everywhere in the neighborhood of z_0 .

✓ Examples:

(1). $f(z) = \frac{1}{z}$ is analytic at each non-zero point in the finite complex plane.

(2). $f(z) = |z|^2$ is not analytic at any non-zero point because it is not differentiable at any non-zero complex number.

(3). $f(z) = \bar{z}$ is nowhere analytic because it is nowhere differentiable.

✓ Remark:

(1). Analytic functions are also known as regular or holomorphic functions.

(2). A function f is analytic everywhere in domain D iff it is analytic at each point of domain D .

(3). A function f is analytic everywhere in domain D then f is entire function in D .

❖ **CAUCHY-RIEMANN EQUATIONS[C-R EQUATION]:**

- ✓ If $u(x, y)$ and $v(x, y)$ are real single-valued functions of x and y such that $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ and $\frac{\partial v}{\partial y}$ are continuous in the region R , then

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial u}{\partial y} = - \frac{\partial v}{\partial x}$$

are known as Cauchy-Riemann Equations.

❖ **NECESSARY AND SUFFICIENT CONDITIONS FOR $f(z)$ TO BE ANALYTIC:**

- ✓ The necessary and sufficient conditions for the function $f(z) = u(x, y) + iv(x, y)$ to be analytic in a region R are

(1). $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ and $\frac{\partial v}{\partial y}$ are continuous functions of x and y in the region R .

(2). $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ & $\frac{\partial u}{\partial y} = - \frac{\partial v}{\partial x}$

i.e. Cauchy-Riemann equations are satisfied.

- ✓ Remark:

(1). C.R. equations are necessary condition for differentiability but not sufficient.

(2). If $f(z) = u(x, y) + iv(x, y)$ is an analytic function, then $u(x, y)$ and $v(x, y)$ are conjugate functions.

(3). If a function is differentiable $\Rightarrow$ function satisfies C.R. equation. If a function does not satisfy C.R. equation $\Rightarrow$ function is not differentiable.

(4). If function is differentiable at point (x_0, y_0) then derivative at z_0 is given by

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0). \quad (\text{Cartesian form})$$

$$f'(z_0) = e^{-i\theta}(u_r(r, \theta) + iv_r(r, \theta)). \quad (\text{Polar form})$$

❖ **CAUCHY-RIEMANN EQUATIONS IN POLAR FORM:**

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} \quad \text{and} \quad \frac{\partial v}{\partial r} = - \frac{1}{r} \frac{\partial u}{\partial \theta}$$

METHOD – 7: ANALITICITY OF COMPLEX NUMBER

C	1	State necessary and sufficient condition for function to be analytic and prove that necessary condition.	
H	2	Show that the function $f(z) = xy + iy$ is continuous everywhere but is not analytic.	
C	3	Check Whether $f(z) = \bar{z}$ is analytic or not. Answer: Nowhere analytic	
H	4	Check Whether $f(z) = 2x + ixy^2$ is analytic or not at any point. Answer: Nowhere analytic	
C	5	State the necessary condition for $f(z)$ to be analytic. For what values of z is the function $f(z) = 3x^2 + iy^2$ analytic ? Answer: Except the line $y = 3x$ function is nowhere analytic	
H	6	Separate in real and imaginary part of $f(z) = e^z$. Also prove that it is analytic everywhere. Answer: $\text{Ref}(z) = e^x \cos y$; $\text{Imf}(z) = e^x \sin y$	
H	7	Check Whether $f(z) = e^{\bar{z}}$ is analytic or not at any point. Answer: Nowhere analytic	
C	8	Consider $f(z) = z^n$ for integer n . Verify C-R equation & find its derivative. Answer: Analytic everywhere & $f'(z) = n z^{n-1}$	
H	9	Is $f(z) = \sqrt{r}e^{\frac{i\theta}{2}}$ analytic? ($r > 0, -\pi < \theta < \pi$) Answer: Yes	
T	10	Check Whether $f(z) = z^{\frac{5}{2}}$ is analytic or not. Answer: Analytic everywhere	
H	11	Check Whether the following functions are analytic or not at any point, (i) $f(z) = 3x + y + i(3y - x)$ (ii) $f(z) = z^{\frac{3}{2}}$. Answer: Both functions are Analytic everywhere	W 2019 (4)

H	12	Check Whether the function $f(z) = \sin z$ is analytic or not. If analytic find its derivative. Answer: Analytic everywhere & $f'(z) = \cos z$	
T	13	Examine the analyticity of $\sinh z$. Answer: Everywhere analytic	
C	14	Show that for the function $f(z) = \begin{cases} \frac{\bar{z}^2}{z} & ; z \neq 0 \\ 0 & ; z = 0 \end{cases}$ is not differentiable at $z = 0$ even though Cauchy Reimann equation are satisfied at $z = 0$.	
C	15	Show that the function $f(z) = \sqrt{ xy }$ satisfies the C-R equation at the origin but $f'(0)$ fails to exist.	
C	16	If w is any function of x and y with differential co efficient of 1st and 2nd order then prove that, $4 \frac{\partial^2 w}{\partial z \partial \bar{z}} = \frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2}$. Also prove that, $\frac{\partial^2 w}{\partial z \partial \bar{z}} = 0$; when w is analytic function.	
C	17	If $f(z)$ is regular function of z , then prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) f(z) ^2 = 4 f'(z) ^2$.	
C	18	If $f(z)$ is analytic function, prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \log f'(z) = 0$.	
C	19	If $f(z)$ is analytic function, prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \operatorname{Re}(f(z)) ^2 = 2 f'(z) ^2$.	W 2019 (4)

❖ HARMONIC FUNCTIONS:

- ✓ A real valued function $\phi(x, y)$ is said to be harmonic function in domain D if

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0. \text{ (Laplace Equation)}$$

- ✓ All second order partial derivative $\phi_{xx}, \phi_{xy}, \phi_{yx}, \phi_{yy}$ are continuous.

❖ **THEOREM:**

- ✓ If $f(z) = u + iv$ is analytic in domain D then u & v are harmonic function in D .

❖ **HARMONIC CONJUGATE:**

- ✓ Let, $u(x, y)$ and $v(x, y)$ are harmonic function and they satisfy C.R. equations in certain domain D then $v(x, y)$ is harmonic conjugate of $u(x, y)$.

❖ **THEOREM:**

- ✓ If $f(z) = u + iv$ is analytic in D iff $v(x, y)$ is harmonic conjugate of $u(x, y)$.

❖ **REMARK:**

- ✓ If $f(z) = u + iv$ is analytic function then $v(x, y)$ is harmonic conjugate of $u(x, y)$ but $u(x, y)$ is not harmonic conjugate of $v(x, y)$. $-u(x, y)$ is harmonic conjugate of $v(x, y)$.

❖ **MILNE-THOMSON'S METHOD:**

- ✓ This method determines the analytic function $f(z)$ when either u or v is given.
- ✓ We know that $z = x + iy$ and $\bar{z} = x - iy$.

$$\text{So, } x = \frac{z + \bar{z}}{2} \text{ \& } y = \frac{z - \bar{z}}{2i}$$

- ✓ Now, $f(z) = u(x, y) + iv(x, y) = u\left(\frac{z+\bar{z}}{2}, \frac{z-\bar{z}}{2i}\right) + iv\left(\frac{z+\bar{z}}{2}, \frac{z-\bar{z}}{2i}\right)$

- ✓ Putting $\bar{z} = z$, we get $f(z) = u(z, 0) + iv(z, 0)$

Which is same as $f(z) = u(x, y) + iv(x, y)$ if we replace x by z and y by 0 .

- ✓ Now, $f(z) = u + iv$

$$\Rightarrow f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y} \quad (\text{By C. R. equations})$$

- ✓ Replacing x by z and y by 0 , we get

$$\Rightarrow f'(z) = u_x(z, 0) - i u_y(z, 0)$$

- ✓ Integrating both the sides, with respect to z , we get

$$\Rightarrow f(z) = \int u_x(z, 0) dz - i \int u_y(z, 0) dz + c.$$

METHOD – 8: TO FIND HARMONIC FUNCTION

C	1	Define: Harmonic Function. Show that $u = x \sin x \cosh y - y \cos x \sinh y$ is harmonic.	
H	2	Define: Harmonic Function. Show that $u = \frac{x}{x^2 + y^2}$ is harmonic function for $\mathbb{R}^2 - (0,0)$.	
C	3	Define: Harmonic Function. Show that $u(x, y) = x^2 - y^2$ is harmonic. Find the corresponding analytic function $f(z) = u(x, y) + iv(x, y)$. Answer: $f(z) = z^2 + c$	
H	4	Define: Harmonic Function. Show that $u(x, y) = x^2 - y^2 + x$ is harmonic. Find the corresponding analytic function $f(z) = u(x, y) + iv(x, y)$. Answer: $f(z) = z^2 + z + c$	
C	5	Show that the function $u = x^3 - 3xy^2$ is harmonic and find the corresponding analytic function. Answer: $f(z) = z^3 + c$	
H	6	Show that $u(x, y) = 2x - x^3 + 3xy^2$ is harmonic in some domain and find a harmonic conjugate $v(x, y)$. Answer: $v(x, y) = 2y - 3x^2y + y^3 + c$	
H	7	Determine a and b such that $u = ax^3 + bxy$ is harmonic and find Conjugate harmonic. Answer: $a = 0 ; b \in \mathbb{C}$	
T	8	Define Harmonic Function. Show that the function $u(x, y) = e^x \cos y$ is harmonic. Determine its harmonic conjugate $v(x, y)$ and the analytic function $f(z) = u + iv$. Answer: $v(x, y) = e^x \sin y ; f(z) = e^x \cos y + i e^x \sin y$	
H	9	Define Harmonic Function. Show that $u(x, y) = \sinh x \sin y$ is harmonic function, find its harmonic conjugate $v(x, y)$. Answer: $v(x, y) = -\cosh x \cos y$	W 2019 (7)

C	10	Determine the analytic function whose real part is $e^x(x \cos y - y \sin y)$. Answer: $f(z) = z e^z + c$	
H	11	Determine the analytic function whose real part is $e^{2x}(x \cos 2y - y \sin 2y)$. Answer: $f(z) = z e^{2z} + c$	
T	12	Show that $u(x, y) = e^{x^2-y^2} \cos(2xy)$ is harmonic everywhere. Also find a conjugate harmonic for $u(x, y)$. Answer: $v(x, y) = e^{x^2-y^2} \sin(2xy)$	
C	13	Find the analytic function $f(z) = u + iv$, if $u - v = e^x(\cos y - \sin y)$. Answer: $f(z) = e^z + c$	
H	14	Find the all analytic function $f(z) = u + iv$, if $u - v = (x - y)(x^2 + 4xy + y^2)$. Answer: $f(z) = -i z^3 + c$	
T	15	Find the analytic function $f(z)$ if $u - v = \frac{e^y - \cos x \sin x}{\cos hy - \cos x}$ with $f\left(\frac{\pi}{2}\right) = \frac{3-i}{2}$. Answer: $f(z) = \cot \frac{z}{2} + \frac{1}{1+i}$	

❖ DEFINITIONS:

✓ Conformal Mapping:

Suppose two curves c_1 and c_2 intersect at point P in Z -plane and the corresponding two curves c_1' and c_2' at P' in the W -plane.

If the angle of intersection of the curves at P is same as the angle of intersection of the curve P' in both magnitude and sense, then the transformation is said to be Conformal.

✓ **Note:** If function is analytic, then mapping is conformal.

✓ Fixed Point (Invariant Point):

Fixed points of mapping $w = f(z)$ are points that are mapped onto themselves are “kept fixed” under the mapping.

Bilinear linear transformations having α and β as fixed point is given by

$$w = \frac{\gamma z - \alpha\beta}{z - (\alpha + \beta) + \gamma} ; \text{ for various values of } \gamma.$$

✓ **Critical Point:**

The point where $f'(z) = 0$ are called Critical Point. At critical point, mapping is not conformal.

✓ **Ordinary Point:**

The point where $f'(z) \neq 0$ is called Ordinary Point.

METHOD – 9: FIXED POINT, CRITICAL POINT, ORDINARY POINT

C	1	Find Fixed point of bilinear trans. (I) $w = \frac{z}{2-z}$ (II) $w = \frac{(2+i)z-2}{z+i}$ (III) $w = \frac{z-1}{z+1}$ (S – 16) Answer: (I) $\alpha = 0, \beta = 1$ (II) $\alpha = 1+i, \beta = 1-i$ (III) $\alpha = i, \beta = -i$	
H	2	Find fixed point of $w = \frac{z+1}{z}$ and verify your result. Answer: $\alpha = \frac{1}{2} + \frac{\sqrt{5}}{2}$, $\beta = \frac{1}{2} - \frac{\sqrt{5}}{2}$	
C	3	Define Critical point & find critical point of the $w = z + z^2$. Answer: $z = -\frac{1}{2}$	
C	4	What does conformal mapping mean? At what points is the mapping by $w = z^2 + \frac{1}{z^2}$ not conformal ? Answer: $z = \pm 1, \pm i$	
H	5	Check whether $w = \frac{1}{z}$ is conformal mapping or not. Answer: Yes	

METHOD – 10: ELEMENTARY TRANSFORMATION

C	1	Find and sketch the image of the region $ z > 1$ under the transformation $w = 4z$. Answer: $ w > 4$	
H	2	Determine & sketch the image of $ z = 1$ under the transformation $w = z + i$. Answer: $ w - i = 1$	S 2020 (4)
H	3	Show that the region in the z - plane given by $x > 0, 0 < y < 2$ has the image $-1 < u < 1, v > 0$ in the w -plane under the transformation $w = iz + 1$.	
H	4	Find the image of infinite strip $0 \leq x \leq 1$ under the transformation $w = iz + 1$. Sketch the region in ω – plane. Answer: $0 \leq v \leq 1$	
C	5	Determine and sketch the image of region $0 \leq x \leq 1, 0 \leq y \leq \pi$ under the transformation $w = e^z$. Answer: $-1 \leq u \leq e, 0 \leq v \leq e$	
C	6	Find & sketch (plot) the image of the region $x \geq 1$ under the transformation $w = \frac{1}{z}$. Answer: $\left w - \frac{1}{2}\right \leq \frac{1}{2}$	
H	7	Find the image of infinite strip (i) $\frac{1}{4} \leq y \leq \frac{1}{2}$ (ii) $0 < y < \frac{1}{2}$ under the transformation $w = \frac{1}{z}$. Show the region graphically. Answer: (i) Region between the circles $w + 2i \leq 2$ & $w + i \geq 1$ (ii) Intersection of region $v < 0$ & $w + i > 1$	W 2019 (7)
T	8	Find image of critical $ z = 1$ under transformation $w = f(z) = \frac{z - i}{1 - iz}$ & find fixed points. Answer: $v = 0$	

C	9	Find the image in the w – plane of the circle $ z - 3 = 2$ in the z – plane under the inversion mapping $w = \frac{1}{z}$. Answer: $\left w - \frac{3}{5}\right = \frac{4}{5\sqrt{2}}$	
T	10	Find the image of $ z - 3i = 3$ under the mapping $w = \frac{1}{z}$. Answer: $v = \frac{1}{6}$	
H	11	Explain translation, rotation and magnification transformation. Find the image of the $ z - 1 = 1$ under transformation $w = \frac{1}{z}$. Answer: $u = \frac{1}{2}$	
C	12	Find the image of region bounded by $1 \leq r \leq 2$ & $\frac{\pi}{6} \leq \theta \leq \frac{\pi}{3}$ in the z – Plane under the transformation $w = z^2$. Show the region graphically. Answer: $1 \leq r' \leq 4$ & $\frac{\pi}{3} \leq \theta' \leq \frac{2\pi}{3}$	
C	13	Determine the points where $w = z + \frac{1}{z}$ is not conformal mapping. Also find image of circle $ z = 2$ under the transformation $w = z + \frac{1}{z}$. Answer: $\frac{u^2}{25} + \frac{v^2}{9} = \frac{1}{4}$	
T	14	Determine the region of w -plane into which the region bounded by $x = 1, y = 1, x + y = 1$ is mapped by the transformation $w = z^2$. Answer: Region bounded by $u^2 = 1 - 2v$; $v^2 = 4 - 4u$; $v^2 = 4 + 4u$	

❖ **BILINEAR TRANSFORMATION / LINEAR FRACTIONAL / MOBIUS TRANSFORMATION:**

- ✓ A transformation of the form $w = \frac{az+b}{cz+d}$; Where a, b, c, d are complex constants and $ad - bc \neq 0$ is called a Bilinear Transformation.

❖ **DETERMINATION OF BILINEAR TRANSFORMATION:**

- ✓ If w_1, w_2, w_3 are the respective images of distinct points z_1, z_2, z_3 then

$$\frac{(w - w_1)(w_2 - w_3)}{(w - w_3)(w_2 - w_1)} = \frac{(z - z_1)(z_2 - z_3)}{(z - z_3)(z_2 - z_1)}$$

METHOD - 11: BILINEAR TRANSFORMATION

C	1	Determine the Mobius transformation which maps $z = \infty, i, 0$ into $w = 0, i, \infty$. Answer: $w = -\frac{1}{z}$	S 2020 (4)
H	2	Determine the Mobius transformation which maps $z_1 = 0, z_2 = 1, z_3 = \infty$ into $w_1 = -1, w_2 = -i, w_3 = 1$. Answer: $w = -\left(\frac{1 + iz}{1 - iz}\right)$	W 2019 (3)
H	3	Determine bilinear transformation which maps point $0, i, 1$ into $i, -1, \infty$. Answer: $w = i\left(\frac{1 + z}{1 - z}\right)$	
H	4	Determine the Linear Fractional Transformation that maps $z_1 = 0, z_2 = 1, z_3 = \infty$ onto $w_1 = -1, w_2 = -i, w_3 = 1$ respectively. Answer: $w = \frac{z - i}{z + i}$	
C	5	Find the bilinear transformation which maps $z = 1, i, -1$ into the points $w = 0, 1, \infty$. Answer: $w = \frac{i - iz}{z + 1}$	
T	6	Find the bilinear transformation which maps $z = 2, 1, 0$ into the points $w = 1, 0, i$. Answer: $w = \frac{2z - 2}{z - iz + 2i}$	
T	7	Find bilinear transformation, which maps the points $1, -1, \infty$ onto the points $1 + i, 1 - i, 1$. Also find fixed point. Answer: $w = 1 + \frac{i}{z}$; Fixed pt. $= \frac{1 \pm \sqrt{1 + 4i}}{2}$	

C	8	Define a Linear Fractional Transformation. Find the bilinear transformation that maps the points $z_1 = -1, z_2 = 0, z_3 = 1$ on to $w_1 = -i, w_2 = 1, w_3 = i$ respectively. Also find w for $z = \infty$. Answer: $w = \frac{i - z}{i + z}$; when $z \rightarrow \infty, w = -1$	
H	9	Find the Bilinear transformation which maps $z = 1, i, -1$ into $w = 2, i, -2$. Answer: $w = \frac{2i - 6z}{iz - 3}$	
C	10	Find the Bilinear transformation which maps $z = 1, i, -1$ into $w = i, 0, 1$. Answer: $w = \frac{(1 + i) - (1 - i)z}{2}$	



UNIT-2 » COMPLEX INTEGRAL, SEQUENCE AND SERIES

❖ INTRODUCTION:

Integrals of complex valued function of a complex variable are defined on curves in the complex plane, rather than on interval of real line.

✓ **Continuous arc:**

The set of points (x, y) defined by $x = f(t)$, $y = g(t)$, with parameter t in the interval (a, b) , define a continuous arc provided f and g are continuous functions.

✓ **Smooth arc:**

If f and g are differentiable on arc $a \leq t \leq b$ and non-zero on open interval $a < t < b$ is called smooth arc.

✓ **Simple Curve/Simple arc/Jordan arc:**

A curve which does not intersect with itself. i.e. if $z(t_1) \neq z(t_2)$ when $t_1 \neq t_2$.

✓ **Simple Closed Curve:**

A simple curve C except for the fact $z(b) = z(a)$; where a & b are end points of interval.

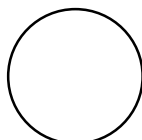


Fig. A



Fig. B

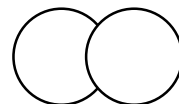


Fig. C

✓ **Contour:**

A contour or piecewise smooth arc, is an arc consisting of a finite number of smooth arcs join end to end.

If only initial and final values are same, a contour is called **Simple closed contour**.

✓ **Simply connected:**

A Simply connected domain D is simple closed path in D encloses only points of D .

Examples: A open disk, ellipse or any simple closed curve.

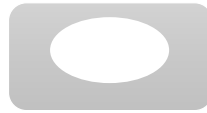
✓ **Multiply connected:**

A domain that is not simply connected is called **multiply connected**.

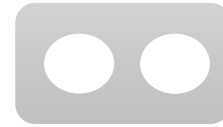
Examples: An annulus is multiply connected.



Simply connected



Doubly Connected



Triply Connected

❖ LINE INTEGRAL IN COMPLEX PLANE:

- ✓ A line integral of a complex function $f(z)$ along the curve C is denoted by $\int_C f(z) dz$.
- ✓ Note that, if C is closed path, then line integral of $f(z)$ is denoted by $\oint_C f(z) dz$.
- $\oint_C f(z) dz$ is known as Contour integral.
- ✓ Properties of Line Integral:

(1). Linearity

$$\int_C [k_1 f(z) + k_2 g(z)] dz = k_1 \int_C f(z) dz + k_2 \int_C g(z) dz$$

(2). Reversing the sense of integration

$$\int_a^b f(z) dz = - \int_b^a f(z) dz$$

(3). Partition of Path

$$\int_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz ; \text{ where } C = C_1 \cup C_2$$

METHOD – 1: LINE INTEGRAL

C	1	Evaluate $\int_C \operatorname{Re}(z) dz$ where C is the shortest path from $(1, 1)$ to $(3, 2)$. Answer: $4 + 2i$	
---	----------	--	--

T	2	Evaluate $\int_0^{2+i} z^2 dz$ along the line joining the points (0,0) and (2,1). Answer: $\frac{2}{3} + \frac{11}{3}i$	
C	3	Evaluate $\int_C (x^2 - iy^2) dz$, along the parabola $y = 2x^2$ from (1,2) to (2,8). Answer: $\frac{511}{3} - \frac{49}{5}i$	S 2020 (3)
H	4	Evaluate $\int_C \bar{z} dz$ from $z = 1 - i$ to $z = 3 + 2i$ along the straight line. Answer: $\frac{11}{2} + 5i$	
H	5	Evaluate $\int_C z^2 dz$ where C is line joins point (0,0) to (4,2). Answer: $\frac{16}{3} + \frac{88}{3}i$	
C	6	Evaluate $\int_0^{2+4i} \operatorname{Re}(z) dz$ along the curve $z = t + it^2$. Answer: $2 + i\frac{16}{3}$	W 2019 (3)
H	7	Find the value of $\int_0^{2+i} (\bar{z})^2 dz$, along the real axis from 0 to 2 and then vertically from 2 to $2 + i$. Answer: $\frac{14 + 11i}{3}$	
T	8	Evaluate $\int_0^{2+i} (\bar{z})^2 dz$, along the shortest path joining the end points. Answer: $\frac{10 - 5i}{3}$	

C	9	Evaluate $\int_c (x - y + ix^2)dz$, Where c is a straight line $z = 0$ to $z = 1 + i$. Answer: $-\frac{1}{3} + \frac{i}{3}$	W 2019 (3)
H	10	Evaluate $\int_c \{(x + y)dx + x^2ydy\}$ along $y = x^2$ having $(0,0), (3,9)$ end points. Answer: $\frac{513}{2}$	
H	11	Evaluate $\int_c \bar{z} dz$, where C is along the sides of triangle having vertices $z = 0, 1, i$. Answer: i	
C	12	Evaluate $\int_c z^2 dz$, Where c is the path joining the points $1 + i$ and $2 + 4i$ along the parabola $x^2 = y$. Answer: $-\frac{86}{3} - 6i$	
H	13	Evaluate $\int_c z^2 dz$, Where c is taken along triangle in $z -$ plane having vertices $z = \pm i, -1$ taken in counter clockwise sense. Answer: $\frac{2i}{3}$	
T	14	Evaluate $\int_c z ^2 dz$ When c is the boundary of the square with vertices $(0,0), (1,0), (1,1), (0,1)$. Answer: $-1 + i$	

C	15	Evaluate $\int_c f(z) dz$. Where $f(z)$ is defined by $f(z) = \begin{cases} 1 & ; y < 0 \\ 4y & ; y > 0 \end{cases}$. c is the arc from $z = -1 - i$ to $z = 1 + i$ along the curve $y = x^3$. Answer: $2 + 3i$	
C	16	Find the value of integral $\int_c \bar{z} dz$ where c is the right – hand half $z = 2e^{i\theta} ; \left(-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2} \right)$ of the circle $z = 2$, from $z = -2i$ to $z = 2i$. Answer: $4\pi i$	

❖ **MAXIMUM MODULUS THEOREM:**

- ✓ If $f(z)$ is analytic inside and on a simple closed curve C , then maximum value of $|f(z)|$ occurs on C , unless $f(z)$ must be constant.

METHOD – 2: MAXIMUM MODULUS THEOREM

C	1	Find an upper bound for the absolute value of the integral $\int_c e^z dz$, where c is the line segment joining the points $(0,0)$ and $(1, 2\sqrt{2})$. Answer: $3e$	
C	2	Without using integration, show that $\left \int_c \frac{e^z}{z+1} dz \right \leq \frac{8\pi e^4}{3} ; C: z = 4$.	
H	3	Find an upper bound for the absolute value of the integral $\int_c \frac{dz}{z^2 + 1}$, where c is the arc of a circle $z = 2$ that lies in the first quadrant. Answer: $\frac{\pi}{3}$	

T	4	Find an upper bound for the absolute value of the integral $\int_c z^2 dz$, where c is the straight line segment from 0 to $1 + i$. Answer: $2\sqrt{2}$	
---	---	--	--

❖ CAUCHY'S INTEGRAL THEOREM (CAUCHY GOURSAT'S THEOREM):

- ✓ If $f(z)$ is an analytic function in a simply connected domain D and $f'(z)$ is continuous at each point within and on a simple closed curve C in D , then

$$\int_C f(z) dz = 0$$

❖ LIOUVILLE'S THEOREM:

- ✓ If $f(z)$ is an analytic and bounded function for all z in the entire complex plane, then $f(z)$ is constant.

METHOD – 3: CAUCHY INTEGRAL THEOREM

C	1	Evaluate $\int_C e^{z^2} dz$, where C is any closed contour. Justify your answer. Answer: 0	
H	2	If C is any simple closed contour, in either direction, then show that $\int_C \exp(z^3) dz = 0$. Answer: 0	
C	3	Evaluate $\oint_C (z^2 + 3) dz$, where C is any closed contour. Justify your answer. Answer: 0	

H	4	Evaluate $\oint_C (z^2 - 2z - 3)dz$, where C is the circle $ z = 2$. Answer: 0	
C	5	Evaluate $\oint_C \frac{e^{-z}}{z+1} dz$, where $C : z = \frac{1}{2}$. Answer: 0	S 2020 (3)
T	6	Evaluate $\oint_C \frac{z}{z-3} dz$, where C is the unit circle $ z = 1$. Answer: 0	
C	7	Evaluate $\oint_C \frac{z+4}{z^2+2z+5} dz$, where C is the circle $ z+1 = 1$. Answer: 0	S 2020 (3)

❖ CAUCHY'S INTEGRAL FORMULA:

- ✓ If $f(z)$ is an analytic within and on a simple closed curve C and z_0 is any point interior to C, then

$$\oint_C \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0)$$

the integration being taken counterclockwise.

- ✓ In general,

$$\oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz = \frac{2\pi i}{n!} f^{(n)}(z_0)$$

METHOD – 4: CAUCHY INTEGRAL FORMULA

C	1	Evaluate $\oint_C \frac{z^2 - 4z + 4}{z + i} dz$, where C is $ z = 2$. Answer: $(-8 + 6i)\pi$	
T	2	Evaluate $\oint_C \frac{\sin 3z}{z + \frac{\pi}{2}} dz$, where C is the circle is $ z = 5$. Answer: $2\pi i$	
C	3	Evaluate the following: (i) $\int_C \frac{z+3}{z-1} dz$; Where, c is the circle (a) $ z = 2$ (b) $ z = \frac{1}{2}$. (ii) $\int_C \frac{\sin z}{\left(z - \frac{\pi}{4}\right)^3} dz$; Where, c is the circle $ z = 1$. Answer: (i) (a) $8\pi i$; (b) 0 (ii) $-\frac{\pi i}{\sqrt{2}}$	W 2019 (7)
C	4	Evaluate $\oint_C \frac{e^{2z}}{(z - \ln 1.5)^3} dz$, where C is the square with vertices $(1,0), (0,1), (-1,0), (0,-1)$. Answer: $9\pi i$	
H	5	Evaluate $\oint_C \frac{1}{z^2} dz$, where C: $ z = 1$. Answer: 0	S 2020 (3)
H	6	Evaluate $\oint_C \frac{5z+7}{z^2+2z-3} dz$, where C is $ z-2 = 2$. Answer: $6\pi i$	
C	7	Evaluate $\oint_C \frac{dz}{z^2+1}$, where C is $ z+i = 1$, counterclockwise. Answer: $-\pi$	

H	8	Evaluate $\oint_C \frac{2z+6}{z^2+4} dz$, where C is $ z-i = 2$. Answer: $(3+2i)\pi$	
H	9	Using Cauchy Integral formula, Evaluate $\int_C \frac{3z^2+2}{(z-2)(z^2+4)} dz$, where C is $ z-2 = 2$. Answer: $\frac{7\pi i}{2}$	
C	10	Evaluate $\oint_C \frac{1}{(z-1)^2(z-3)} dz$, where C is $ z = 2$. Answer: $-\frac{\pi}{2}i$	
T	11	Evaluate $\oint_C \frac{z-1}{(z+1)^2(z-2)} dz$, where C is $ z-i = 2$. Answer: $-\frac{2\pi i}{9}$	
H	12	Evaluate $\int_C \frac{z}{z^2+1} dz$, where c is the circle (i) $\left z+\frac{1}{2}\right = 2$ (ii) $ z+i = 1$. Answer: (I) $2\pi i$; (II) πi	
C	13	Evaluate $\int_C \frac{1+z^2}{1-z^2} dz$, where c is unit circle centred at (I) $z = -1$ (II) $z = i$. Answer: (I) $2\pi i$; (II) 0	
H	14	Evaluate $\int_C \frac{z^2+1}{z^2-1} dz$, if c is unit circle of unit radius with centre (I) $z = 1$ (II) $z = -1$. Answer: (I) $2\pi i$; (II) $-2\pi i$	

T	15	State Cauchy – Integral theorem. Evaluate $\oint_C \left(\frac{3}{z-i} - \frac{6}{(z-i)^2} \right) dz$, Where C: $ z = 2$. Answer: $6\pi i$	
H	16	Evaluate $\int_{C: z =2} \frac{dz}{z^3(z+4)}$. Answer: $\frac{\pi i}{32}$	
C	17	Evaluate $\oint_C \frac{e^z}{z(1-z)^3} dz$, where, C is (I) $ z = \frac{1}{2}$ (II) $ z-1 = \frac{1}{2}$. Answer: (I) $2\pi i$; (II) $-\pi i$	

❖ **SEQUENCES:**

- ✓ A sequence is obtained by assigning to each positive integer n , a number z_n , called a term of the sequence, and is written $z_1, z_2, z_3, \dots$ OR $\{z_1, z_2, z_3, \dots\}$ OR $\{z_n\}$.

❖ **CONVERGENT SEQUENCE:**

- ✓ A sequence is called convergent, if $\lim_{n \rightarrow \infty} z_n = c$.
- ✓ In other words, for every $\epsilon > 0$, we can find an N such that $|z_n - c| < \epsilon$ for all $n > N$.
- ✓ A Divergent sequence is one that does not converge.
- ✓ Example:

(1). $\left\{ \frac{i^n}{n} \right\} = \left\{ i, \frac{-1}{2}, \frac{-i}{3}, \frac{1}{4}, \dots \right\}$ is convergent with limit 0.

(2). $\{i^n\} = \{i, -1, -i, 1, \dots\}$ is divergent.

❖ **THEOREM:**

- ✓ A sequence $z_1, z_2, z_3, \dots, z_n, \dots$ of complex numbers $z_n = x_n + iy_n$ ($n = 1, 2, 3, \dots$) converges to $c = a + ib$ if and only if the sequence of the real parts $x_1, x_2, x_3, \dots$ converges to a and the sequence of the imaginary parts $y_1, y_2, y_3, \dots$ converges to b .
- ✓ Example:

- The sequence $\{z_n = (1 - \frac{1}{n^2}) + i(2 + \frac{4}{n})\}$ converges to $1 + 2i$ as $\{a_n = 1 - \frac{1}{n^2}\}$ converges to 1 and $\{b_n = 2 - \frac{4}{n}\}$ converges to 2.

METHOD – 5: CONVERGENCE OF A SEQUENCE

C	1	Is sequence $\{i^{2n}\}$ convergent or divergent ? Answer: Divergent	
C	2	Is sequence $\{1 + \frac{2in}{n+1}\}$ convergent or divergent ? Answer: Convergent	
H	3	Is sequence $\{\frac{1 + 3n^2i}{2n^2 - n}\}$ convergent or divergent ? Answer: Convergent	
H	4	Is sequence $\{\frac{1 + 2n^2}{n^2} - \frac{n-1}{n}i\}$ convergent or divergent ? Answer: Convergent	
C	5	Is sequence $\{e^{\frac{n\pi i}{3}}\}$ convergent or divergent ? Answer: Divergent	
T	6	Is sequence $\{-(i^{4n})\}$ convergent or divergent ? Answer: Convergent	
C	7	Is sequence $\{\sin(ni)\}$ convergent or divergent ? Answer: Divergent	

❖ SERIES:

- ✓ Given a sequence $z_1, z_2, z_3, \dots$, we may form the sequence of the sums

$$s_1 = z_1 ; s_2 = z_1 + z_2 ; s_3 = z_1 + z_2 + z_3, \dots$$

$$\text{In general, } s_n = z_1 + z_2 + z_3 + \dots + z_n$$

- ✓ Here, s_n is called the n th partial sum of the series.

❖ **CONVERGENT SERIES:**

- ✓ A series is called convergent, if $\lim_{n \rightarrow \infty} s_n = s$.
- ✓ We may write, $s = \sum_{m=1}^{\infty} z_m$. Here, s is called the sum of series.

❖ **THEOREM:**

- ✓ A series of complex numbers with $z_m = x_m + iy_m$ converges to $s = u + iv$ if and only if

(1). $x_1 + x_2 + x_3 + \dots$ converges to u .

(2). $y_1 + y_2 + y_3 + \dots$ converges to v .

❖ **THEOREM:**

- ✓ If a series $z_1 + z_2 + z_3 + \dots$ converges, then $\lim_{m \rightarrow \infty} z_m = 0$.
- ✓ If $\lim_{m \rightarrow \infty} z_m \neq 0$, then series $z_1 + z_2 + z_3 + \dots$ diverges.
- ✓ **Note:** If $\lim_{m \rightarrow \infty} z_m = 0$, then series $z_1 + z_2 + z_3 + \dots$ may not converges.

❖ **ABSOLUTE CONVERGENCE:**

- ✓ A series $z_1 + z_2 + z_3 + \dots$ is called absolutely convergent, if the series of the absolute values of the terms

$$\sum_{m=1}^{\infty} |z_m| = |z_1| + |z_2| + |z_3| + \dots$$

is convergent.

- ✓ If a series is absolutely convergent, it is convergent.

❖ **CONDITIONALLY CONVERGENCE:**

- ✓ If $z_1 + z_2 + z_3 + \dots$ converges but $|z_1| + |z_2| + |z_3| + \dots$ diverges, then the series $z_1 + z_2 + z_3 + \dots$ is called, more precisely, conditionally convergent.

❖ **COMPARISON TEST:**

- ✓ If a series $z_1 + z_2 + z_3 + \dots$ is given and find convergent series $b_1 + b_2 + b_3 + \dots$ with non - negative real terms such that $|z_1| \leq b_1, |z_2| \leq b_2, |z_3| \leq b_3, \dots$ then the given series converges, even absolutely.

❖ **GEOMETRIC SERIES:**

- ✓ The geometric series

$$\sum_{m=0}^{\infty} q^m = 1 + q + q^2 + \dots$$

(1). Diverges, if $|q| \geq 1$.(2). Converges, if $|q| < 1$.

$$\text{Also, } \sum_{m=0}^{\infty} q^m = \frac{1}{1-q}.$$

❖ **RATIO TEST:**

- ✓ If a series
- $z_1 + z_2 + z_3 + \dots$
- is given with
- $z_n \neq 0$
- is such that
- $\lim_{n \rightarrow \infty} \left| \frac{z_{n+1}}{z_n} \right| = L$
- .

(1). If $L < 1$, the series converges absolutely.(2). If $L > 1$, the series diverges.(3). If $L = 1$, test fails.❖ **ROOT TEST:**

- ✓ If a series
- $z_1 + z_2 + z_3 + \dots$
- is given with
- $z_n \neq 0$
- is such that
- $\lim_{n \rightarrow \infty} \sqrt[n]{|z_n|} = L$
- .

(1). If $L < 1$, the series converges absolutely.(2). If $L > 1$, the series diverges.(3). If $L = 1$, test fails.**METHOD – 6: CONVERGENCE OF SERIES**

C	1	Check series $\sum_{n=0}^{\infty} \frac{(-i)^n}{\ln n}$ is convergent or divergent. Answer: Divergent	
H	2	Check series $\sum_{n=0}^{\infty} \frac{i^n}{n^2 - i}$ is convergent or divergent. Answer: Convergent	

H	3	Check series $\sum_{n=0}^{\infty} \frac{(\pi + \pi i)^{2n+1}}{(2n+1)!}$ is convergent or divergent. Answer: Convergent	
C	4	Check series $\sum_{n=0}^{\infty} \frac{(-1)^n(1+i)^{2n}}{(2n)!}$ is convergent or divergent. Answer: Convergent	

❖ POWER SERIES

- ✓ A series of the form

$$\sum_{n=0}^{\infty} a_n(z - z_0)^n = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots + a_n(z - z_0)^n + \cdots$$

in which the coefficients $a_n \in \mathbb{C}$; $n = 0, 1, \dots$ and z_0 is a fixed point in the complex z -plane is called a power series in powers of $(z - z_0)$ or about z_0 or centered at z_0 .

❖ CONVERGENCE OF A POWER SERIES IN A DISK

- ✓ The series converges everywhere in a circular disk $|z - z_0| < R$ and diverges everywhere in the disk $|z - z_0| > R$.
- ✓ Here, R is called the radius of convergence and the circle $|z - z_0| = R$ is called the circle of convergence.

❖ RADIUS OF CONVERGENCE

- ✓ For, power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ Radius of convergence R is defined as

$$R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| \quad \text{OR} \quad R = \lim_{n \rightarrow \infty} |a_n|^{-\frac{1}{n}}$$

METHOD - 7: RADIUS OF CONVERGENCE

H	1	Find the radius of convergence of $\sum_{n=0}^{\infty} (n + 2i)^n z^n$. Answer: $R = 0$	
---	---	---	--

C	2	Find the radius of convergence of $\sum_{n=1}^{\infty} \left(\frac{6n+1}{2n+5}\right)^2 (z-2i)^n$. Answer: R = 1	
H	3	Find the radius of convergence of $\sum_{n=1}^{\infty} (6+8i)^n z^n$. Answer: R = $\frac{1}{6+8i}$	
T	4	Find the radius of convergence of $\sum_{n=0}^{\infty} z^n$. Answer: R = 1	
H	5	Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{2^n}{n!} z^n$. Answer: R = ∞	
C	6	Find the radius of convergence of $\sum_{n=2}^{\infty} \frac{z^n}{n(\log n)^3}$. Answer: R = 1	
C	7	Find the radius of convergence of $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{n^2} z^n$. Answer: R = $\frac{1}{e}$	S 2020 (4)

❖ FUNCTION GIVEN BY POWER SERIES

- ✓ If any power series $\sum_{n=0}^{\infty} a_n(z-z_0)^n$ has a non – zero radius of convergence R, its sum is a function of z, say f(z). We write,

$$f(z) = \sum_{n=0}^{\infty} a_n(z-z_0)^n = a_0 + a_1(z-z_0) + a_2(z-z_0)^2 + \dots \quad |z-z_0| < R.$$

- ✓ Here, f(z) is a called function given by power series.
- ✓ **Note:** A function cannot be represented by two different power series at same centre.

❖ **TAYLOR'S SERIES:**

- ✓ Let $f(z)$ be analytic everywhere inside a circle C with centre at z_0 and radius R . then at each point Z inside C , we have

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \frac{f''(z_0)}{2!}(z - z_0)^2 + \dots + \frac{f^n(z_0)}{n!}(z - z_0)^n + \dots$$

❖ **MACLAURIN'S SERIES:**

- ✓ If we take $z_0 = 0$, in Taylor's series reduces to

$$f(z) = f(0) + f'(0)z + \frac{f''(0)}{2!}z^2 + \dots + \frac{f^n(0)}{n!}z^n + \dots$$

❖ **UNIFORM CONVERGENCE**

- ✓ A series

$$\sum_{m=0}^{\infty} f_m(z) = f_0(z) + f_1(z) + f_2(z) + \dots$$

With sum $s(z)$ is called uniformly convergent in a region G if for every $\epsilon > 0$.

- ✓ We can find an $N = N(\epsilon)$ not depending on z , such that $|s(z) - s_n(z)| < \epsilon$, $\forall n > N(\epsilon), z \in G$.
- ✓ Uniformity of convergence is thus a property that always refers to an infinite set in the z -plane, that is, a set consisting of infinitely many points.

❖ **UNIFORM CONVERGENCE OF POWER SERIES**

- ✓ A power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ with a non-zero radius of convergence R is uniformly convergent in every circular disk $|z - z_0| \leq r$ of radius $r < R$.

METHOD - 8: TAYLOR'S SERIES AND MACLAURIN'S SERIES

C	1	Derive the Taylor's series representation in $\frac{1}{1-z} = \sum_{n=0}^{\infty} \frac{(z-i)^n}{(1-i)^{n+1}} ; \text{ where } z-i < \sqrt{2}.$	
H	2	Obtain the Taylor's series $f(z) = \sin z$ in power of $\left(z - \frac{\pi}{4}\right)$.	

H	3	Determine Taylor's series expansion of $\frac{2z^2 + 9z + 5}{(z - 3)(z + 2)^2}$ with center at $z_0 = 1$.	
C	4	Find Maclaurin series representation of $f(z) = \sin z$ in the region $ z < \infty$.	
C	5	Find the series expression for $f(z) = \tan^{-1} z$ at $z = 0$.	
H	6	Expand $f(z) = \frac{z - 1}{z + 1}$ as a Taylor's series about the point $z = 0$.	



UNIT-3 » LAURENT'S SERIES AND RESIDUES

❖ LAURENT'S SERIES:

- ✓ If $f(z)$ is analytic within and on the ring (annulus) shaped region R bounded by two concentric circles C_1 and C_2 of radii R_1 and R_2 ($R_2 < R_1$) resp. having center at the point $z = z_0$, then for all z in R , $f(z)$ is uniquely represented by a convergent Laurent's series given by

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} a_{-n} (z - z_0)^{-n}$$

$$\text{Where, } a_n = \frac{1}{2\pi i} \int_{C_1} \frac{f(t)}{(t - z_0)^{n+1}} dt \quad \& \quad a_{-n} = \frac{1}{2\pi i} \int_{C_2} \frac{f(t)}{(t - z_0)^{-n+1}} dt$$

Here, $\sum_{n=1}^{\infty} a_{-n} (z - z_0)^{-n}$ is known Principal Part of Laurent's series.

❖ GEOMETRIC SERIES:

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n \quad (|z| < 1)$$

$$\frac{1}{1+z} = \sum_{n=0}^{\infty} (-1)^n z^n \quad (|z| < 1)$$

METHOD - 1: LAURENT'S SERIES

C	1	Show that when $0 < z - 1 < 2$, $\frac{z}{(z-1)(z-3)} = \frac{-1}{2(z-1)} - 3 \sum_{n=0}^{\infty} \frac{(z-1)^n}{2^{n+2}}$	
C	2	Find the Laurent's expansion in power of z that represent $f(z) = \frac{1}{z(z-1)}$ for domain (a) $0 < z < 1$ (b) $0 < z-1 < 1$.	
H	3	Find the Laurent's expansion of $f(z) = \frac{7z-2}{(z+1)z(z-2)}$ in the region $1 < z+1 < 3$.	
C	4	Expand $f(z) = \frac{1}{(z-1)(z+2)}$ valid for the region (a) $ z < 1$ (b) $1 < z < 2$ (c) $ z > 2$.	W 2019 (7)

H	5	Expand $f(z) = \frac{1}{(z-1)(z-2)}$ in the region (a) $ z < 1$ (b) $1 < z < 2$.	
H	6	Expand $f(z) = -\frac{1}{(z-1)(z-2)}$ in the region (a) $ z < 1$ (b) $1 < z < 2$ (c) $ z > 2$.	
H	7	Expand $f(z) = \frac{1}{(z+2)(z+4)}$ for the region (a) $ z < 2$ (b) $2 < z < 4$ (c) $ z > 4$.	S 2020 (7)
C	8	Expand $\frac{1}{z(z-1)(z-2)}$ for the regions (a) $ z < 1$ (b) $1 < z < 2$ (c) $ z > 2$.	S 2020 (7)
H	9	Find the Laurent's series expansion of $f(z) = \frac{1}{6-z-z^2}$ in the domain (a) $ z < 2$ (b) $2 < z < 3$ (c) $ z > 3$.	
T	10	Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in Laurent's series in the interval $1 < z < 3$.	
T	11	Write the two Laurent series expansion in powers of z that represent the function $f(z) = \frac{1}{z^2(1-z)}$ in certain domains, also specify domains.	

❖ DEFINITION:

✓ **Singular point:**

A point z_0 is a singular point if a function $f(z)$ is not analytic at z_0 but is analytic at some points of each neighborhood of z_0 .

✓ **Isolated point:**

A singular point z_0 of $f(z)$ is said to be isolated point if there is a neighborhood of z_0 which contains no singular points of $f(z)$ except z_0 .

✓ In other words, $f(z)$ is analytic in some deleted neighborhood, $0 < |z - z_0| < \varepsilon$.

$$\text{e.g. } f(z) = \frac{z^2 + 1}{(z-1)(z-2)} \text{ has two isolated point } z = 1 \text{ \& } z = 2.$$

✓ **Poles:**

If principal part of Laurent's series has finite number of terms,

$$\text{i. e. } f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n + \frac{b_1}{z - z_0} + \frac{b_2}{(z - z_0)^2} + \dots + \frac{b_n}{(z - z_0)^n}$$

then the singularity $z = z_0$ is said to be pole of order n .

✓ If $b_1 \neq 0$ and $b_2 = b_3 = \dots = b_n = 0$, then

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n + \frac{b_1}{z - z_0}$$

the singularity $z = z_0$ is said to be pole of order 1 or a simple pole.

❖ **TYPES OF SINGULARITIES:**✓ **Removable singularity:**

If in Laurent's series expansion, If the principal part is zero,

$$\text{i. e. } f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n + 0$$

then the singularity $z = z_0$ is said to be removable singularity.

In other words, $f(z)$ is not defined at $z = z_0$ but $\lim_{z \rightarrow z_0} f(z)$ exists.

$$\text{e. g. } f(z) = \frac{\sin z}{z} \text{ is undefined at } z = 0 \text{ but } \lim_{z \rightarrow 0} \frac{\sin z}{z} = 1.$$

So, $z = 0$ is a removable singularity.

✓ **Essential singularity:**

If in the Laurent's series expansion, the principal part contains an infinite number of terms, then the singularity $z = z_0$ is said to be an essential singularity.

$$\text{e. g. } f(z) = \sin \frac{1}{z} \text{ has an essential singularity at } z = 0.$$

$$\because \sin \frac{1}{z} = \frac{1}{z} - \frac{1}{3!} \frac{1}{z^3} + \frac{1}{5!} \frac{1}{z^5} + \dots$$

❖ RESIDUE OF A FUNCTION:

- ✓ If $f(z)$ has a pole at the point $z = z_0$ then the coefficient b_1 of the term $\frac{1}{z-z_0}$ in the Laurent's series expansion of $f(z)$ at $z = z_0$ is called the residue of $f(z)$ at $z = z_0$.
- ✓ Residue of $f(z)$ at $z = z_0$ is denoted by $\text{Res}_{z=z_0} f(z)$.

❖ TECHNIQUE TO FIND RESIDUE:

(1). If $f(z)$ has a simple pole at $z = z_0$, then

$$\text{Res}(f(z_0)) = \lim_{z \rightarrow z_0} (z - z_0)f(z)$$

(2). If $f(z) = \frac{P(z)}{Q(z)}$ has a simple pole at $z = z_0$, then

$$\text{Res}(f(z_0)) = \frac{P(z_0)}{Q'(z_0)}$$

(3). If $f(z)$ has a pole of order n at $z = z_0$, then

$$\text{Res}(f(z_0)) = \frac{1}{(n-1)!} \lim_{z \rightarrow z_0} \frac{d^{(n-1)}}{dz^{(n-1)}} [(z - z_0)^n f(z)]$$

METHOD – 2: RESIDUES OF FUNCTION

C	1	Classify the poles of $f(z) = \frac{1}{z^2 - z^6}$. Answer: <table border="1" style="margin: 10px auto; border-collapse: collapse; text-align: center;"> <tr> <td>z_0</td><td>0</td><td>± 1</td><td>$\pm i$</td></tr> <tr> <td>n</td><td>2</td><td>1</td><td>1</td></tr> <tr> <td>Type of pole</td><td>Double</td><td>Simple</td><td>Simple</td></tr> </table>	z_0	0	± 1	$\pm i$	n	2	1	1	Type of pole	Double	Simple	Simple	
z_0	0	± 1	$\pm i$												
n	2	1	1												
Type of pole	Double	Simple	Simple												
C	2	Find the order of pole of the point $z = 0$ for the function $f(z) = \frac{\sin z}{z^4}$. Answer: pole of order $n = 3$													
C	3	Define residue at simple pole and find the sum of residues of the function $f(z) = \frac{\sin z}{z \cos z}$ at its poles inside the circle $ z = 2$. Answer: 0													

C	4	Find the residue at $z = 0$ of $f(z) = \frac{1 - e^z}{z^3}$. Answer: $-\frac{1}{2}$									
H	5	Expand Laurent series of $f(z) = \frac{1 - e^z}{z}$ at $z = 0$ and identify the singularity. Answer: $z = 0$ is a removable singularity.	W 2019 (3)								
H	6	Find the residue at $z = 0$ of $f(z) = z \cos \frac{1}{z}$. Answer: Residue $= -\frac{1}{2}$									
H	7	Show that the singular point of the function $f(z) = \frac{1 - \cosh z}{z^3}$ is a pole. Determine the order m of that pole and corresponding residue. Answer: $m = 1$; $\text{Res}(f(z), 0) = -\frac{1}{2}$									
C	8	For $f(z) = \frac{1}{(z - 1)^2(z - 3)}$, find the residue at $z = 1$. Answer: $\text{Res}(f(z), 1) = -\frac{1}{4}$	S 2020 (4)								
H	9	Define residue at simple pole. Find the residues at each of its poles of $f(z) = \frac{z^2 - 2z}{(z + 1)^2(z^2 + 4)}$ in the finite z - plane. Answer: <table border="1" data-bbox="331 1496 933 1686"> <tr> <td>Pole</td><td>$-2i$</td><td>$2i$</td><td>-1</td></tr> <tr> <td>Residue</td><td>$\frac{7 - i}{25}$</td><td>$\frac{7 + i}{25}$</td><td>$\frac{14}{25}$</td></tr> </table>	Pole	$-2i$	$2i$	-1	Residue	$\frac{7 - i}{25}$	$\frac{7 + i}{25}$	$\frac{14}{25}$	
Pole	$-2i$	$2i$	-1								
Residue	$\frac{7 - i}{25}$	$\frac{7 + i}{25}$	$\frac{14}{25}$								

❖ CAUCHY'S RESIDUE THEOREM:

- ✓ If $f(z)$ is analytic in a closed curve C except at a finite number of singular points within C , then

$$\int_C f(z) dz = 2\pi i \text{ (sum of the residue at the singular points)}$$

METHOD – 3: CAUCHY'S RESIDUE THEOREM

C	1	State Cauchy's residue theorem. Evaluate $\int_C \frac{5z-2}{z(z-1)} dz$, where C is the circle $ z = 2$. Answer: $10\pi i$	
H	2	Evaluate $\int_C \frac{2z^2+3}{z(z+1)(z+2)} dz$; C: $ z = 1.6$ using Residue theorem. Answer: $-7\pi i$	
H	3	Evaluate $\oint_C \frac{\cos \pi z^2}{(z-1)(z-2)} dz$, where C is the circle $ z = 3$. Answer: $4\pi i$	
C	4	Using residue theorem, Evaluate $\oint_C \frac{e^z+z}{z^3-z} dz$, Where C: $ z = \frac{\pi}{2}$. Answer: $\pi i \left(e - 2 + \frac{1}{e} \right)$	
H	5	Using residue theorem, Evaluate $\oint_C \frac{z^2 \sin z}{4z^2-1} dz$, C: $ z = 2$. Answer: $\frac{\pi i}{2} \sin \frac{1}{2}$	
C	6	Determine the poles of the function $f(z) = \frac{z^2}{(z-1)^2(z+2)}$ and residue at each pole. Hence evaluate $\int_C f(z) dz$, where C: $ z = 3$. Answer: $2\pi i$	

C	7	Find the value of the integral $\int_C \frac{2z^2 + 2}{(z-1)(z^2+9)} dz$ taken counterclockwise around the circle C: $ z-2 = 2$. Answer: $\frac{4}{5}\pi i$	
H	8	Evaluate $\oint \frac{z^3 - z^2 + z - 1}{z^3 + 4z} dz$ where c is the circle $ z = 3$ and $ z = 1$ using Cauchy's residue theorem. Answer: $-2\pi i$; $-\frac{\pi i}{2}$	
H	9	Evaluate $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)} dz$, where C is the circle $ z = 3$. Answer: $4\pi i(\pi + 1)$	
C	10	Evaluate $\int_C \frac{dz}{(z^2+1)^2}$, where C: $ z+i = 1$. Answer: $-\frac{\pi}{2}$	
C	11	Evaluate $\oint_C e^{\frac{3}{z}} dz$, where C is $ z = 1$. Answer: $6\pi i$	
H	12	Use residues to evaluate the integrals of the function $\frac{\exp(-z)}{z^2}$ around the circle $ z = 3$ in the positive sense. Answer: $-2\pi i$	
C	13	Evaluate $\oint_C \frac{e^{2z}}{(z+1)^3} dz$ where C : $4x^2 + 9y^2 = 16$ using residue theorem. Answer: $4\pi i e^{-2}$	S 2020 (4)
T	14	Evaluate $\oint_C \tan z dz$, where C is the circle $ z = 2$. Answer: $-4\pi i$	

T	15	Evaluate $\oint_C \frac{dz}{\sinh 2z}$, Where C: $z = 2$. Answer: πi	
---	----	--	--

❖ INTEGRATION AROUND THE UNIT CIRCLE:

- ✓ An integral of the type $\int_0^{2\pi} F(\cos \theta, \sin \theta) d\theta$, where $F(\cos \theta, \sin \theta)$ is a rational function of $\cos \theta$ and $\sin \theta$ can be evaluated by taking $z = e^{i\theta}$.

$$\text{Here, } z = e^{i\theta}$$

$$\Rightarrow dz = i e^{i\theta} d\theta$$

$$\Rightarrow \frac{dz}{i e^{i\theta}} = d\theta$$

$$\Rightarrow d\theta = \frac{dz}{i z}$$

$$\text{Now, } \cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$= \frac{1}{2} \left(z + \frac{1}{z} \right) = \frac{1}{2} \left(\frac{z^2 + 1}{z} \right)$$

$$\text{Also, } \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$= \frac{1}{2i} \left(z - \frac{1}{z} \right) = \frac{1}{2i} \left(\frac{z^2 - 1}{z} \right)$$

- ✓ Now, the given integral takes the form $\int_C f(z) dz$, where $f(z)$ is a rational function of z and c is the unit circle $|z| = 1$.
- ✓ This complex integral can be evaluated using the residue theorem.

❖ INTEGRATION AROUND A SMALL SEMI-CIRCLE(IMPROPER INTEGRALS OF RATIONAL FUNCTIONS):

- ✓ Let, $f(x) = \frac{P(x)}{Q(x)}$; where $P(x)$ and $Q(x)$ are polynomials of degree m and n respectively.
- ✓ If $Q(x) \neq 0$; for all real x and $n \geq m + 2$, the Cauchy principal value of the integral is

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{P(x)}{Q(x)} dx = 2\pi i \sum_{j=1}^k \text{Res}_{z=z_j} \frac{P(z)}{Q(z)}$$

Where, z_j are the poles of $\frac{P(z)}{Q(z)}$ that lie in the upper half plane.

❖ **IMPROPER INTEGRALS INVOLVING TRIGONOMETRIC FUNCTIONS:**

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{P(x)}{Q(x)} \cos \alpha x dx = -2\pi \sum_{j=1}^k \text{Im} \left[\text{Res}_{z=z_j} \frac{P(z)}{Q(z)} \right]$$

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{P(x)}{Q(x)} \sin \alpha x dx = 2\pi \sum_{j=1}^k \text{Re} \left[\text{Res}_{z=z_j} \frac{P(z)}{Q(z)} \right]$$

METHOD – 4: CONTOUR INTEGRATION BY USING RESIDUE THEOREM

C	1	Using the residue theorem, evaluate $\int_0^{2\pi} \frac{d\theta}{5 + 3 \sin \theta}$. Answer: $\frac{\pi}{2}$	
H	2	Using the residue theorem, evaluate $\int_0^{2\pi} \frac{4 d\theta}{5 + 4 \sin \theta}$. Answer: $\frac{8\pi}{3}$	
H	3	Using the residue theorem, evaluate $\int_0^{2\pi} \frac{d\theta}{(2 + \cos \theta)^2}$. Answer: $\frac{4\pi}{3\sqrt{3}}$	
T	4	Using the residue theorem, evaluate $\int_0^{2\pi} \frac{d\theta}{(2 + \cos \theta)}$. Answer: $\frac{2\pi}{\sqrt{3}}$	

H	5	Evaluate $\int_0^{2\pi} \frac{d\theta}{3 - 2 \cos \theta + \sin \theta}$. Answer: π	
C	6	Evaluate $\int_0^{\pi} \frac{d\theta}{17 - 8 \cos \theta}$, by integrating around a unit circle. Answer: $\frac{\pi}{15}$	
T	7	Evaluate the Cauchy's principle value of $\int_{-\infty}^{\infty} \frac{dx}{(x^2 + 1)(x^2 + 9)}$. Answer: $\frac{\pi}{12}$	
C	8	Use residues to evaluate $\int_0^{\infty} \frac{x^2 dx}{(x^2 + 1)(x^2 + 4)}$. Answer: $\frac{\pi}{6}$	
H	9	Using contour Integration show that $\int_0^{\infty} \frac{dx}{1 + x^4} = \frac{\pi}{2\sqrt{2}}$.	
H	10	Using residue theorem, evaluate $\int_0^{\infty} \frac{dx}{(x^2 + 1)^2}$. Answer: $\frac{\pi}{4}$	W 2019 (7)
T	11	Let $a > b > 0$. Prove that $\int_{-\infty}^{\infty} \frac{\cos x dx}{(x^2 + a^2)(x^2 + b^2)} = \frac{\pi}{a^2 - b^2} \left(\frac{e^{-b}}{b} - \frac{e^{-a}}{a} \right)$.	
C	12	Using the theory of residue, evaluate $\int_{-\infty}^{\infty} \frac{\cos x}{x^2 + 1} dx$. Answer: $\frac{\pi}{e}$	

H	13	Evaluate $\int_{-\infty}^{\infty} \frac{x \cos x}{x^2 + 1} dx$.	
		Answer: 0	
T	14	Evaluate $\int_0^{\infty} \frac{x \sin x}{x^2 + 9} dx$ using residue.	
		Answer: $\frac{\pi}{2e^3}$	
T	15	Evaluate $\int_0^{\infty} \frac{x \cos x}{x^2 + 9} dx$ using residue.	
		Answer: 0	



UNIT-4 » FIRST ORDER PARTIAL DIFFERENTIAL EQUATION

❖ INTRODUCTION

- ✓ A partial differential equation is a mathematical equation involving two or more independent variables, unknown function and its partial derivative with respect to independent variables.
- ✓ Partial differential equations are used to formulate the problems containing functions of several variables, such as propagation of heat or sound, fluid flow, electrodynamics etc.

❖ DEFINITION: PARTIAL DIFFERENTIAL EQUATION:

- ✓ An equation which involves function of two or more variables and partial derivatives of that function then it is called Partial Differential Equation.

e.g. $\frac{\partial y}{\partial x} + \frac{\partial y}{\partial t} = 0.$

❖ DEFINITION: ORDER OF DIFFERENTIAL EQUATION:

- ✓ The order of highest derivative which appears in differential equation is “Order of D.E”.

e.g. $\left(\frac{\partial y}{\partial x}\right)^2 + \frac{\partial y}{\partial t} + 5y = 0$ has order 1.

❖ DEFINITION: DEGREE OF DIFFERENTIAL EQUATION:

- ✓ When a D.E. is in a polynomial form of derivatives, the highest power of highest order derivative occurring in D.E. is called a “Degree Of D.E.”.

e.g. $\left(\frac{\partial y}{\partial x}\right)^2 + \frac{\partial y}{\partial t} + 5y = 0$ has degree 2 .

❖ NOTATION:

- ✓ Suppose $z = f(x, y)$. For that , we shall use $\frac{\partial z}{\partial x} = p$, $\frac{\partial z}{\partial y} = q$, $\frac{\partial^2 z}{\partial x^2} = r$, $\frac{\partial^2 z}{\partial x \partial y} = s$, $\frac{\partial^2 z}{\partial y^2} = t$.

❖ FORMATION OF PARTIAL DIFFERENTIAL EQUATION:

- ✓ By Eliminating Arbitrary Constants

Consider the function $f(x, y, z, a, b) = 0$. Where, a & b are arbitrary constants.

Step 1: $f(x, y, z, a, b) = 0$(1)

Step 2: $f_x(x, y, z, a, b) = 0$(2) and $f_y(x, y, z, a, b) = 0$(3)

Step 3: Eliminate a & b from eq. (1), eq. (2) & eq. (3).

We get partial differential equation of the form $F(x, y, z, p, q) = 0$

✓ By Eliminating Arbitrary Functions

➤ Type 1: Consider, the function $(u, v) = 0$; u and v are functions of x and y

Step 1: Let, $u = F(v)$.

Step 2: Find u_x & u_y .

Step 3: Eliminate the function F from u_x & u_y .

Note: In such case, for elimination of function, substitution method is used.

➤ Type 2: Consider, the function $z = f(x, y)$

Step 1: Find z_x & z_y .

Step 2: Eliminate the function f from z_x & z_y .

Note: In such case, for elimination of function, division of z_x & z_y is used.

METHOD – 1: EXAMPLE ON FORMATION OF PARTIAL DIFFERENTIAL EQUATION

H	1	Form the partial differential equation $z = ax + by + ct$. Answer: $z = px + qy + t \frac{\partial z}{\partial t}$	
H	2	Derive partial differential equation by eliminating the arbitrary constants 'a' and 'b' from $z = ax + by + ab$. Answer: $z = px + qy + pq$	S 2020 (3)
C	3	Form the partial differential equation $z = (x - 2)^2 + (y - 3)^2$. Answer: $4z = p^2 + q^2$	
C	4	Form the partial differential equation for the equation $(x - a)(y - b) - z^2 = x^2 + y^2$. Answer: $4(x + pz)(y + qz) - z^2 = x^2 + y^2$	

H	5	Eliminate the function f from the relation $f(xy + z^2, x + y + z) = 0$. Answer: $\frac{p + 1}{q + 1} = \frac{y + 2zp}{x + 2zq}$	
C	6	Form the partial differential equation of $f(x + y + z, x^2 + y^2 + z^2) = 0$. Answer: $\frac{p + 1}{q + 1} = \frac{x + zp}{y + zq}$	
T	7	Form a partial differential equation by eliminating the arbitrary function ϕ from $\phi(x + y + z, x^2 + y^2 - z^2) = 0$. Answer: $(y + z)p - (x + z)q = x - y$	
H	8	Form the partial differential equation $f(x^2 - y^2, xyz) = 0$. Answer: $\frac{yz + xyp}{xz + xyq} = -\frac{x}{y}$	
C	9	Form the partial differential equations by eliminating the arbitrary function from $f(x^2 + y^2, z - xy) = 0$. Answer: $\frac{x}{y} = \frac{p - y}{q - x}$	
C	10	Form partial differential equation by eliminating the arbitrary function from $xyz = \Phi(x + y + z)$. Answer: $\frac{p + 1}{q + 1} = \frac{yz + xyp}{xz + xyq}$	
H	11	Form the partial differential equation by eliminating arbitrary function ϕ from $z = \phi\left(\frac{y}{x}\right)$. Answer: $\frac{p}{q} = -\frac{y}{x}$	S 2020 (4)
H	12	Form the partial differential equation by eliminating the arbitrary function from $z = f(x^2 - y^2)$. Answer: $\frac{p}{q} = -\frac{x}{y}$	
T	13	Form the partial differential equation of $z = xy + f(x^2 + y^2)$. Answer: $\frac{p - y}{q - x} = \frac{x}{y}$	
C	14	Form partial differential equation of $z = f(x + iy) + g(x - iy)$. Answer: $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$	

❖ LAGRANGE'S DIFFERENTIAL EQUATION:

- ✓ A partial differential equation of the form $Pp + Qq = R$ where P, Q and R are functions of x, y, z , or constant is called lagrange linear equation of the first order.

❖ METHOD FOR OBTAINING GENERAL SOLUTION OF $Pp + Qq = R$:

Step-1: From the A.E. $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$.

Step-2: Solve this A.E. by the method of grouping or by the method of multiples or both to get two independent solution $u(x, y, z) = c_1$ and $v(x, y, z) = c_2$.

Step-3: The form $F(u, v) = 0$ or $u = f(v)$ & $v = f(u)$ is the general solution $Pp + Qq = R$

❖ FOLLOWING TWO METHODS WILL BE USED TO SOLVE LAGRANGE'S LINEAR EQUATION

➤ Grouping Method

- This method is applicable only if the third variable z is absent in $\frac{dx}{P} = \frac{dy}{Q}$ or it is possible to eliminate z from $\frac{dx}{P} = \frac{dy}{Q}$.
- Similarly, if the variable x is absent in last two fractions or it is possible to eliminate x from last two fractions $\frac{dy}{Q} = \frac{dz}{R}$, then we can apply grouping method.

➤ Multipliers Method

- In this method, we require two sets of multiplier l, m, n and l', m', n' .
- By appropriate selection multiplier l, m, n (either constants or functions of x, y, z) we may write

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{l dx + m dy + n dz}{lP + mQ + nR} \text{ Such that, } lP + mQ + nR = 0.$$

- This implies $l dx + m dy + n dz = 0$
Solving it we get $u(x, y, z) = c_1 \quad \dots (1)$
- Again we may find another set of multipliers l', m', n' . So that, $l'P + m'Q + n'R = 0$
This gives, $l' dx + m' dy + n' dz = 0$
- Solving it we get $v(x, y, z) = c_2 \quad \dots (2)$

- From (1) and (2), we get the general solution as $F(u, v) = 0$.

METHOD – 2: EXAMPLE ON LAGRANGE'S DIFFERENTIAL EQUATION

C	1	Solve $x^2p + y^2q = (x + y)z$. Answer: $f\left(\frac{x-y}{z}, \frac{1}{y} - \frac{1}{x}\right) = 0$	W 2019 (4)
H	2	Solve $x^2p + y^2q = z^2$. Answer: $f\left(\frac{1}{y} - \frac{1}{x}, \frac{1}{z} - \frac{1}{y}\right) = 0$	S 2020 (3)
C	3	Solve $y^2p - xyq = x(z - 2y)$. Answer: $f(x^2 + y^2, yz - y^2) = 0$	
H	4	Solve $xp + yq = 3z$. Answer: $f\left(\frac{x}{y}, \frac{y^3}{z}\right) = 0$	
H	5	Find the general solution to the P.D.E. $xp + yq = x - y$. Answer: $f\left(\frac{x}{y}, x - y - z\right) = 0$	S 2020 (3)
H	6	Solve $z(xp - yq) = y^2 - x^2$. Answer: $f(xy, (x + y)^2 + z^2) = 0$	W 2019 (4)
C	7	Solve $(z - y)p + (x - z)q = y - x$. Answer: $f(x + y + z, x^2 + y^2 + z^2) = 0$	
H	8	Solve $x(y - z)p + y(z - x)q = z(x - y)$. Answer: $f(x + y + z, xyz) = 0$	
C	9	Solve $(x^2 - y^2 - z^2)p + 2xyq = 2xz$. Answer: $f\left(\frac{y}{z}, \frac{x^2 + y^2 + z^2}{z}\right) = 0$	
T	10	Solve $(y + z)p + (x + z)q = x + y$. Answer: $f\left(\frac{x-y}{y-z}, (x + y + z)(x - y)^2\right) = 0$	
T	11	Solve $x^2(y - z)p + y^2(z - x)q = z^2(x - y)$. Answer: $f\left(xyz, \frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right) = 0$	

C	12	Solve $(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$.	
		Answer: $f\left(\frac{x-y}{y-z}, \frac{y-z}{z-x}\right) = 0$	

❖ NON LINEAR PARTIAL DIFFERENTIAL EQUATION OF FIRST ORDER:

- ✓ A partial differential equation in which p & q occur in more than one order is known as Non Linear Partial Differential Equation.

- ✓ **Type 1:** Equation Of the form $f(p, q) = 0$.

Step 1: Substitute $p = a$ & $q = b$.

Step 2: Convert $b = g(a)$.

Step 3: Complete Solution : $z = ax + by + c \Rightarrow z = ax + g(a)y + c$

- ✓ **Type 2:** Equation Of the form $f(x, p) = g(y, q)$.

Step 1: $f(x, p) = g(y, q) = a$

Step 2: Solving equations for p & q. Assume $p = F(x)$ & $q = G(y)$.

Step 3: Complete Solution : $z = \int F(x) dx + \int G(y) dy + b$.

- ✓ **Type 3:** Equation Of the form $z = px + qy + f(p, q)$ (Clairaut's form.) W-15

Step 1: Substitute $p = a$ & $q = b$.

Step 2: Complete Solution : $z = ax + by + f(a, b)$.

- ✓ **Type 4:** Equation Of the form $f(z, p, q) = 0$.

Step 1: Assume $q = ap$

Step 2: Solve the Equation in $dz = p dx + q dy$

METHOD – 3: EXAMPLE ON NON-LINEAR PDE

C	1	Solve $p^2 + q^2 = 1$.	
		Answer: $z = ax \pm (\sqrt{1 - a^2})y + c$	

H	2	Solve $\sqrt{p} + \sqrt{q} = 1$. Answer: $z = ax + (1 - \sqrt{a})^2 y + c$	
H	3	Find the complete integral of $q = pq + p^2$. Answer: $z = ax + \frac{a^2}{1-a} y + c; a \neq 1$	
C	4	Solve $p^2 + q^2 = npq$. Answer: $z = ax + \frac{na \pm a\sqrt{n^2 - 4}}{2} y + c$	
T	5	Find the complete integral of $p^2 = q + x$. Answer: $z = \frac{2}{3}(x + a)^{\frac{3}{2}} + ay + b$	
C	6	Solve $p^2 + q^2 = x + y$. Answer: $z = \frac{2}{3}(a + x)^{\frac{3}{2}} + \frac{2}{3}(y - a)^{\frac{3}{2}} + b$	
H	7	Solve $p^3 + q^3 = x + y$. Answer: $z = \frac{3}{4}(x + a)^{\frac{4}{3}} + \frac{3}{4}(y - a)^{\frac{4}{3}} + b$	W 2019 (3)
T	8	Solve $p^2 - q^2 = x - y$. Answer: $z = \frac{2}{3}(a + x)^{\frac{3}{2}} + \frac{2}{3}(a + y)^{\frac{3}{2}} + b$	
H	9	Solve $p - x^2 = q + y^2$. Answer: $z = ax + \frac{x^3}{3} + ay - \frac{y^3}{3} + b$	
C	10	Solve $z = px + qy + p^2 q^2$. Answer: $z = ax + by + a^2 b^2$	
C	11	Find the complete integral of $p^2 = qz$. Answer: $\log z = ax + a^2 y + b$	S 2020 (3)
H	12	Solve $qz = p(1 + q)$. Answer: $\log(az - 1) = x + ay + b$	
C	13	Solve $pq = 4z$. Answer: $az = (x + ay + b)^2$	

❖ CHARPIT'S METHOD:

✓ Consider, $f(x, y, z, p, q) = 0$.

Step 1: Find value of p & q by using the relation

$$\frac{dx}{\frac{\partial f}{\partial p}} = \frac{dy}{\frac{\partial f}{\partial q}} = \frac{dz}{p \frac{\partial f}{\partial p} + q \frac{\partial f}{\partial q}} = \frac{dp}{-\left(\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}\right)} = \frac{dq}{-\left(\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}\right)} \quad (\text{lagrange - Charpit eq}^n)$$

Step 2: Find value of p & q.

Step 3: Complete Solution : $z = \int p \, dx + \int q \, dy + c$.

METHOD - 4: EXAMPLE ON CHARPIT'S METHOD

C	1	Using charpit's method solve $z = px + qy + p^2 + q^2$. Answer: $z = ax + by + a^2 + b^2$	
C	2	Solve $px + qy = pq$ by charpit's method. Answer: $2az = (ax + y)^2 + b$	W 2019 (4)
H	3	Solve by charpit's method $(p^2 + q^2)y = qz$. Answer: $\sqrt{z^2 - a^2y^2} = ax + b$	
T	4	Solve $py = 2yx + \log q$. Answer: $az = ax^2 + a^2x + e^{ay} + ab$	
C	5	Solve $z^2 = p q x y$. Answer: $z = b x^a y^{\frac{1}{a}}$	
C	6	Solve by Charpit's method $yzp^2 - q = 0$. Answer: $z\sqrt{a - y^2} = x + b$	S 2020 (4)
C	7	Solve $(p^2 + q^2)y = qz$. Answer: $\sqrt{z^2 - a^2y^2} = ax + b$	
H	8	Solve by charpit's method $p = (z + qy)^2$. Answer: $zy = ax + z\sqrt{a}\sqrt{y} + b$	
H	9	Solve by charpit's method $z = p^2x - q^2y$. Answer: $\sqrt{1+a}\sqrt{z} = \sqrt{a}\sqrt{x} + \sqrt{y} + b$	
T	10	Solve by charpit's method $pxy + pq + qy = yz$. Answer: $\log(z - ax) = y - a \log(a + y) + b$	



UNIT-5 » HIGHER ORDER PARTIAL DIFFERENTIAL EQUATION

❖ LINEAR PDE WITH CONSTANT CO-EFFICIENT:

- ✓ The n^{th} order linear partial differential equation with constant co-efficient is

$$a_0 \frac{\partial^n z}{\partial x^n} + a_1 \frac{\partial^n z}{\partial x^{n-1} \partial y} + \dots + a_n \frac{\partial^n z}{\partial y^n} = F(x, y) \dots \dots \dots (A)$$

Where, $a_0, a_1, \dots, a_n$ are constants.

❖ NOTATIONS:

- ✓ Replacing $\frac{\partial}{\partial x} = D$ and $\frac{\partial}{\partial y} = D'$ in Eq. (A), it can be written in operator form as below,

$$a_0 D^n z + a_1 D^{n-1} D' z + \dots + a_n D'^n z = F(x, y)$$

$$\Rightarrow [f(D, D')]z = F(x, y)$$

❖ AUXILIARY EQUATION:

- ✓ The auxiliary equation for n^{th} order PDE $a_0 D^n z + a_1 D^{n-1} D' z + \dots + a_n D'^n z = F(x, y)$ is derived by replacing D by m , D' by 1 in $f(D, D')$ and $F(x, y)$ by 0 .

❖ COMPLEMENTARY FUNCTION (C.F. = z_c):

- ✓ A general solution of $[f(D, D')]z = 0$ is called complementary function of $[f(D, D')]z = F(x, y)$.

❖ PARTICULAR INTEGRAL (P.I. = z_p):

- ✓ A particular integral of $[f(D, D')]z = F(x, y)$ is P. I. = $\frac{1}{f(D, D')} F(x, y)$.

❖ GENERAL SOLUTION OF PDE:

- ✓ G. S. = C. F. + P. I. = $z_c + z_p$

❖ METHOD FOR FINDING C.F. OF PARTIAL DIFFERENTIAL EQUATION:

- ✓ Consider, $a_0 D^n z + a_1 D^{n-1} D' z + \dots + a_n D'^n z = F(x, y)$.
- ✓ The Auxiliary equation is $a_0 m^n + a_1 m^{n-1} + \dots + a_n = 0$.
- ✓ Let $m_1, m_2, \dots$ be the roots of auxiliary equation.

Case	Nature of the “n” roots	Complementary Function
1.	$m_1 \neq m_2 \neq m_3 \neq m_4 \neq \dots$	$z_c = f_1(y + m_1x) + f_2(y + m_2x) + f_3(y + m_3x) + \dots$
2.	$m_1 = m_2 = m$ $m_3 \neq m_4 \neq \dots$	$z_c = f_1(y + mx) + x f_2(y + mx) + f_3(y + m_3x) + \dots$
3.	$m_1 = m_2 = m_3 = m$ $m_4 \neq m_5, \dots$	$z_c = f_1(y + mx) + x f_2(y + mx)$ $+ x^2 f_3(y + mx) + f_4(y + m_4x) + \dots$

METHOD – 1: EXAMPLE ON SOLUTION OF HOMO. HIGHER ORDERED PDE

C	1	Solve $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} - 6 \frac{\partial^2 z}{\partial y^2} = 0$. Answer: $z = f_1(y - 2x) + f_2(y + 3x)$	
C	2	Solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. Answer: $u = f_1(y + ix) + f_2(y - ix)$	
H	3	Solve $2r + 5s + 2t = 0$. Answer: $z = f_1\left(y - \frac{1}{2}x\right) + f_2(y - 2x)$	S 2020 (4)
C	4	Solve $\frac{\partial^3 z}{\partial x^3} + 3 \frac{\partial^3 z}{\partial x^2 \partial y} - 4 \frac{\partial^3 z}{\partial y^3} = 0$. Answer: $z = f_1(y - 2x) + x f_2(y - 2x) + f_3(y + x)$	
H	5	Solve $\frac{\partial^3 z}{\partial x^3} - 3 \frac{\partial^3 z}{\partial x^2 \partial y} + 2 \frac{\partial^3 z}{\partial y^3} = 0$. Answer: $z = f_1(y + x) + f_2[y + (1 + \sqrt{3})x] + f_3[y + (1 - \sqrt{3})x]$	

❖ METHOD FOR FINDING PARTICULAR INTEGRAL:

- ✓ For partial differential equation the value of Particular integral can be find by following methods.
 - General Method
 - Short-cut Method

❖ GENERAL METHOD

- ✓ Consider the partial differential equation $f(D, D')z = F(x, y)$
- ✓ Particular integral P. I. = $\frac{1}{f(D, D')} F(x, y)$
- ✓ Suppose, $f(x, y)$ is factorized into n linear factors.

$$\text{P. I.} = \frac{1}{f(D, D')} F(x, y) = \frac{1}{(D - m_1 D')(D - m_2 D') \dots (D - m_n D')} F(x, y)$$

Which can be evaluated by

$$\frac{1}{D - mD'} F(x, y) = \int F(x, c - mx) dx$$

- ✓ Where, c is replaced by $y + mx$ after integration.

❖ SHORTCUT METHOD

- ✓ **Case-1** $F(x, y) = e^{ax+by}$

$$\text{P. I.} = \frac{1}{f(D, D')} e^{ax+by} = \frac{1}{f(a, b)} e^{ax+by}, \text{ if } f(a, b) \neq 0$$

- If $f(a, b) = 0$ then $m = \frac{a}{b}$ is a root of auxiliary equation repeated r times.

$$f(D, D') = \left(D - \frac{a}{b} D'\right)^r g(D, D')$$

$$\text{P. I.} = \frac{1}{\left(D - \frac{a}{b} D'\right)^r g(D, D')} e^{ax+by} = \frac{x^r}{r!} \frac{1}{g(a, b)} e^{ax+by}, g(a, b) \neq 0$$

- ✓ **Case-2** $F(x, y) = \sin(ax + by)$

$$\text{P. I.} = \frac{1}{f(D^2, DD', D'^2)} \sin(ax + by) = \frac{1}{f(-a^2, -ab, -b^2)} \sin(ax + by)$$

Where, $f(-a^2, -ab, -b^2) \neq 0$

- If $f(-a^2, -ab, -b^2) = 0$, then use general method for finding P.I.

- ✓ **Case-3** $F(x, y) = \cos(ax + by)$

$$\text{P. I.} = \frac{1}{f(D^2, DD', D'^2)} \cos(ax + by) = \frac{1}{f(-a^2, -ab, -b^2)} \cos(ax + by)$$

Where, $f(-a^2, -ab, -b^2) \neq 0$

➤ If $f(-a^2, -ab, -b^2) = 0$, then use general method for finding P.I.

✓ **Case-4** $F(x, y) = x^m y^n$

$$\text{P.I.} = \frac{1}{f(D, D')} x^m y^n = [f(D, D')]^{-1} x^m y^n$$

➤ Expand $[f(D, D')]^{-1}$ by using binomial expansion according to the following rules:

○ If $n < m$, expand in power of $\frac{D'}{D}$.

○ If $m < n$, expand in power of $\frac{D}{D'}$.

✓ **Case-5** $f(x, y) = e^{ax+by} V(x, y)$

$$\text{P.I.} = \frac{1}{f(D, D')} e^{ax+by} V(x, y) = e^{ax+by} \frac{1}{f(D+a, D'+b)} V(x, y)$$

METHOD - 2: EXAMPLE ON SOLUTION OF NON-HOMO. HIGHER ORDERED PDE

C	1	Solve $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 3e^{x+y}$. Answer: $z = f_1(y-x) + \frac{3}{2}e^{x+y}$	
C	2	Solve $\frac{\partial^2 z}{\partial x^2} - 2\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = e^{x+4y}$. Answer: $z = f_1(y+x) + x f_2(y+x) + \frac{1}{9}e^{x+4y}$	
H	3	Solve $\frac{\partial^2 z}{\partial x^2} + 2\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = e^{2x+3y}$. Answer: $z = f_1(y-x) + x f_2(y-x) + \frac{1}{25}e^{2x+3y}$	W 2019 (3)
H	4	Solve $\frac{\partial^3 z}{\partial x^3} - 2\frac{\partial^3 z}{\partial x^2 \partial y} = 2e^{2x}$. Answer: $z = \phi_1(y) + x\phi_2(y) + \phi_3(y+2x) + \frac{1}{4}e^{2x}$	
H	5	Find complete solution $\frac{\partial^3 z}{\partial x^3} - 3\frac{\partial^3 z}{\partial^2 x \partial y} + 4\frac{\partial^3 z}{\partial y^3} = e^{x+2y}$. Answer: $\phi_1(y-x) + \phi_2(y+2x) + x\phi_3(y+2x) + \frac{e^{x+2y}}{27}$	

T	6	Solve $\frac{\partial^2 z}{\partial x^2} - 4 \frac{\partial^2 z}{\partial x \partial y} + 4 \frac{\partial^2 z}{\partial y^2} = e^{2x+3y}$. Answer: $\phi_1(y + 2x) + x\phi_2(y + 2x) + \frac{e^{2x+3y}}{16}$	
C	7	Solve $(D + D')z = \cos x$. Answer: $z = f(y - x) + \sin x$	
C	8	Solve $\frac{\partial^2 z}{\partial x^2} - 4 \frac{\partial^2 z}{\partial y^2} = \cos 2x \cos 3y$. Answer: $z = f_1(y - 2x) + f_2(y + 2x) + \frac{1}{32} \cos 2x \cos 3y$	
H	9	Solve $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} = \sin x \sin 2y$. Answer: $z = f_1(y) + f_2(y + x) - \frac{1}{6} [\cos(x - 2y) + 3 \cos(x + 2y)]$	
H	10	Solve $(D^2 - 3DD' + 2D'^2)z = \cos(x + 2y)$. Answer: $z = \phi_1(y - 2x) + \phi_2(y - x) - \frac{1}{3} \cos(x + 2y)$	
C	11	Solve $2 \frac{\partial^2 z}{\partial x^2} - 5 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = \sin(2x + y)$. Answer: $z = f_1(2y + x) + f_2(y + 2x) - \frac{1}{3} x \cos(y + 2x)$	
C	12	Solve $\frac{\partial^2 z}{\partial x^2} + 3 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = x + y$. Answer: $z = \phi_1(y - 2x) + \phi_2(y - x) - \frac{x^3}{3} + \frac{x^2 y}{2}$	
H	13	Solve $\frac{\partial^2 z}{\partial x^2} - 4 \frac{\partial^2 z}{\partial y^2} = 3x - 4y$. Answer: $z = f_1(y + 2x) + f_2(y - 2x) + \frac{x^3}{2} - 2x^2 y$	
H	14	Solve $(2D^2 - 5DD' + D'^2)z = 24(y - x)$. Answer: $z = f_1(y + 2x) + f_2\left(y + \frac{x}{2}\right) + \frac{4}{9} (y - x)^3$	S 2020 (3)
C	15	Solve $(D^3 + D^2 D' - DD'^2 - D'^3)z = e^x \cos 2y$ Answer: $z = f_1(y - x) + x f_2(y - x) + f_3(y + x) + \frac{e^x}{25} (\cos 2y + 2 \sin 2y)$	

H	16	Solve $\frac{\partial^2 z}{\partial x^2} - 3 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = (1 + 2x) e^{x+2y}$. Answer: $z = f_1(y + 2x) + f_2(y + x) + \frac{(6x + 11)}{9} e^{x+2y}$	
T	17	Solve $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6 \frac{\partial^2 z}{\partial y^2} = y \sin x$. Answer: $z = f_1(y - 3x) + f_2(y + 2x) - \cos x - y \sin x$	

❖ METHOD OF SEPARATION OF VARIABLES:

- ✓ Follow the below listed step to solve given PDE by using method of Separation of Variables.

Step 1: Let $u(x, y) = X(x) \cdot Y(y)$.

Step 2: Find $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial^2 u}{\partial x^2}, \frac{\partial^2 u}{\partial x \partial y}, \frac{\partial^2 u}{\partial y^2}$ as requirement and substitute in given Partial Differential Eqⁿ.

Step 3: Convert it into Separable Variable equation and equate with constant say k individually.

Step 4: Solve each Ordinary Differential Equation.

Step 5: Put value of $X(x)$ & $Y(y)$ in equation $u(x, y) = X(x) \cdot Y(y)$.

METHOD – 3: EXAMPLE ON SEPARATION OF VARIABLES

H	1	Solve the equation by method of separation of variables $\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y}$, Where $u(0, y) = 8e^{-3y}$. Answer: $u(x, y) = 8 e^{-12x-3y}$	
C	2	Using method of separation of variables $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$. Answer: $u(x, t) = c_1 c_2 e^{kx + \left(\frac{k-1}{2}\right)y}$	W 2019 (3)

H	3	Using method of separation of variables, solve $2u_x = u_t + u$, $u(x, 0) = 4e^{-3x}$. Answer: $u(x, t) = 4e^{-3x - 7t}$	S 2020 (4)
C	4	Using method of separation of variables solve $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 2(x + y)u$. Answer: $u(x, y) = c_1 c_2 e^{x^2 + y^2 + kx - ky}$	S 2020 (4)
H	5	Solve $x \frac{\partial u}{\partial x} - 2y \frac{\partial u}{\partial y} = 0$ using method of separation variables. Answer: $u(x, y) = c_1 c_2 x^k y^{\frac{k}{2}}$	
H	6	Find the general solution of partial differential equation $u_{xx} = 9u_y$, using the method of separation variables. Answer: $u(x, y) = (c_1 e^{\sqrt{k}x} + c_2 e^{-\sqrt{k}x}) c_3 e^{\frac{ky}{9}}$	W 2019 (7)
H	7	Using method of separation of variables solve $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial y} + 2u$. Answer: $u(x, y) = (c_1 e^{\sqrt{k}x} + c_2 e^{-\sqrt{k}x}) c_3 e^{(k-2)y}$	
C	8	Solve two dimensional Laplace's equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, using the method separation of variables. Answer: $u(x, y) = (c_1 e^{\sqrt{k}x} + c_2 e^{-\sqrt{k}x}) (c_3 \cos \sqrt{k}y + c_4 \sin \sqrt{k}y)$	
H	9	Using the method of separation of variables, solve the partial differential equation $\frac{\partial^2 u}{\partial x^2} = 16 \frac{\partial^2 u}{\partial y^2}$. Answer: $u(x, y) = (c_1 e^{\sqrt{k}x} + c_2 e^{-\sqrt{k}x}) (c_3 e^{\frac{\sqrt{k}}{4}y} + c_4 e^{-\frac{\sqrt{k}}{4}y})$	
H	10	Solve the method of separation of variables $\frac{\partial^2 u}{\partial x^2} - 4 \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$. Answer: $u(x, y) = (c_1 e^{(2+\sqrt{4+k})x} + c_2 e^{(2-\sqrt{4+k})x}) c_3 e^{-ky}$	
C	11	Solve $\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$ by the method of separation variable. Answer: $z(x, y) = (c_1 e^{(1+\sqrt{1+k})x} + c_2 e^{(1-\sqrt{1+k})x}) c_3 e^{-ky}$	

❖ CLASSIFICATION OF SECOND ORDER PARTIAL DIFFERENTIAL EQUATION:

- ✓ The general form of a non-homogeneous second order P.D.E.

$$A(x, y) \frac{\partial^2 z}{\partial x^2} + B(x, y) \frac{\partial^2 z}{\partial x \partial y} + C(x, y) \frac{\partial^2 z}{\partial y^2} + f\left(x, y, z, \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}\right) = F(x, y) \dots \dots (1)$$

- ✓ Equation (1) is said to be

Elliptic, If $B^2 - 4AC < 0$

Parabolic, If $B^2 - 4AC = 0$

Hyperbolic, If $B^2 - 4AC > 0$

METHOD – 4: EXAMPLE ON CLASSIFICATION OF 2ND ORDER PDE

C	1	Classify the 2nd order P. D. E. $t \frac{\partial^2 u}{\partial t^2} + 3 \frac{\partial^2 u}{\partial x \partial t} + x \frac{\partial^2 u}{\partial x^2} + 17 \frac{\partial u}{\partial x} = 0$. Answer: Hyperbolic , if $xt < \frac{9}{4}$; Parabolic , if $xt = \frac{9}{4}$ Elliptical , if $xt > \frac{9}{4}$	
H	2	Classify the 2nd order P. D. E. $4 \frac{\partial^2 u}{\partial t^2} - 9 \frac{\partial^2 u}{\partial t \partial x} + 5 \frac{\partial^2 u}{\partial x^2} = 0$. Answer: Hyperbolic	
H	3	Classify the 2nd order P. D. E. $\frac{\partial^2 u}{\partial t^2} + 2 \frac{\partial^2 u}{\partial x \partial t} + 2 \frac{\partial^2 u}{\partial x^2} = 0$. Answer: Elliptic	

❖ SOLUTION OF WAVE EQUATION:

- ✓ The one – dimensional wave equation is

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \dots \dots (1)$$

- ✓ Where, $y(x, t)$ is the deflection of the string.
- ✓ Since the string is fixed at the ends $x = 0, x = l$, we have two boundary conditions.

For, $x = 0, y(0, t) = 0$, for all t . $\dots \dots (2)$

For, $x = l, y(l, t) = 0$, for all t (3)

✓ Initial Conditions

$$y(x, 0) = f(x) \quad \dots (4)$$

$$\left(\frac{\partial y}{\partial t}\right)_{t=0} = g(x) \text{ velocity} \quad \dots (5)$$

✓ Suppose, $y(x, t) = X(x) \cdot T(t)$ is a solution of eq. (1).

$$\Rightarrow \frac{\partial^2 y}{\partial t^2} = X T'', \frac{\partial^2 y}{\partial x^2} = X'' T$$

✓ Substituting in eq. (1)

$$\Rightarrow X T'' = c^2 X'' T$$

$$\Rightarrow \frac{X''}{X} = \frac{1}{c^2} \frac{T''}{T} = k$$

✓ When, $k = 0$

$$X(x) = (c_1 + c_2 x) \text{ and } T(t) = (c_3 + c_4 t)$$

$$y(x, t) = (c_1 + c_2 x)(c_3 + c_4 t)$$

✓ When, k is positive. (i.e. $k = p^2$)

$$\text{Now, } \frac{X''}{X} = p^2$$

$$\Rightarrow D^2 X = X p^2$$

$$\Rightarrow (D^2 - p^2)X = 0$$

Auxiliary Equation,

$$m^2 - p^2 = 0$$

$$\Rightarrow m = +p, -p$$

$$\text{So, } X(x) = c_1 e^{p x} + c_2 e^{-p x}$$

$$\text{Now, } \frac{1}{c^2} \frac{T''}{T} = p^2$$

$$\Rightarrow D^2 T = c^2 T p^2$$

$$\Rightarrow (D^2 - c^2 p^2)T = 0$$

Auxiliary Equation,

$$m^2 - c^2 p^2 = 0$$

$$\Rightarrow m = +c p, -c p$$

$$\text{So, } T(t) = c_3 e^{c p t} + c_4 e^{-c p t}$$

Solution:

$$y(x, t) = X(x) \cdot T(t) = (c_1 e^{p x} + c_2 e^{-p x})(c_3 e^{c p t} + c_4 e^{-c p t})$$

✓ When, k is negative. (i.e. $k = -p^2$)

$$\text{Now, } \frac{X''}{X} = -p^2$$

$$\Rightarrow D^2 X = -X p^2$$

$$\Rightarrow (D^2 + p^2)X = 0$$

Auxiliary Equation,

$$m^2 + p^2 = 0$$

$$\Rightarrow m = +i p, -i p$$

$$\text{So, } X(x) = c_1 \cos p x + c_2 \sin p x$$

$$\text{Now, } \frac{1}{c^2} \frac{T''}{T} = -p^2$$

$$\Rightarrow D^2 T = -c^2 T p^2$$

$$\Rightarrow (D^2 + c^2 p^2)T = 0$$

Auxiliary Equation,

$$m^2 + c^2 p^2 = 0$$

$$\Rightarrow m = +i c p, -i c p$$

$$\text{So, } T(t) = c_3 \cos c p t + c_4 \sin c p t$$

Solution:

$$y(x, t) = X(x) \cdot T(t) = (c_1 \cos p x + c_2 \sin p x)(c_3 \cos c p t + c_4 \sin c p t)$$

- ✓ As string vibration are in harmonic motion y must be a periodic function. So, we can consider only 3rd case. i.e. $k < 0$.

$$y(x, t) = (c_1 \cos p x + c_2 \sin p x)(c_3 \cos c p t + c_4 \sin c p t) \quad \dots \dots (6)$$

- ✓ We have, $y(0, t) = 0$

$$\Rightarrow c_1 (c_3 \cos c p t + c_4 \sin c p t) = 0$$

$$\Rightarrow c_1 = 0$$

$$\text{So, we have } y(x, t) = c_2 \sin p x (c_3 \cos c p t + c_4 \sin c p t) \quad \dots \dots (7)$$

- ✓ We have, $y(l, t) = 0$

$$\Rightarrow c_2 \sin p l (c_3 \cos c p t + c_4 \sin c p t) = 0$$

$$\Rightarrow \sin p l = 0$$

$$\Rightarrow p l = n \pi$$

$$\Rightarrow p = \frac{n \pi}{l} ; n = 1, 2, 3, \dots$$

$$\text{Hence, } y(x, t) = c_2 \sin \left(\frac{n \pi x}{l} \right) \left\{ c_3 \cos \left(\frac{n \pi c t}{l} \right) + c_4 \sin \left(\frac{n \pi c t}{l} \right) \right\}$$

$$\Rightarrow y(x, t) = \sin \left(\frac{n \pi x}{l} \right) \left\{ b_n \cos \left(\frac{n \pi c t}{l} \right) + c_n \sin \left(\frac{n \pi c t}{l} \right) \right\} \quad \text{Where, } b_n = c_2 c_3, c_n = c_2 c_4$$

- ✓ Taking sum of all solution for $n = 1, 2, 3, \dots$

$$\Rightarrow y(x, t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{l}\right) \left\{ b_n \cos\left(\frac{n\pi ct}{l}\right) + c_n \sin\left(\frac{n\pi ct}{l}\right) \right\} \dots \dots (8)$$

From, eq. (4) and eq. (8),

$$\Rightarrow y(x, 0) = f(x) = \sum_{n=1}^{\infty} b_n \cdot \sin\left(\frac{n\pi x}{l}\right) \quad \text{Which is half range sine series.}$$

$$\text{So, } b_n = \frac{2}{l} \int_0^l f(x) \cdot \sin\left(\frac{n\pi x}{l}\right) dx \quad ; n = 1, 2, 3, \dots$$

✓ Differentiating eq. (7),

$$\Rightarrow \frac{\partial y}{\partial t} = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{l}\right) \left\{ -b_n \sin\left(\frac{n\pi ct}{l}\right) + c_n \cos\left(\frac{n\pi ct}{l}\right) \right\} \cdot \left(\frac{n\pi c}{l}\right)$$

$$\Rightarrow \left(\frac{\partial y}{\partial t}\right)_{t=0} = \sum_{n=1}^{\infty} c_n \cdot \sin\left(\frac{n\pi x}{l}\right) \cdot \left(\frac{n\pi c}{l}\right)$$

$$\Rightarrow \left(\frac{\partial y}{\partial t}\right)_{t=0} = \sum_{n=1}^{\infty} \left(c_n \frac{n\pi c}{l}\right) \cdot \sin\left(\frac{n\pi x}{l}\right) \quad \text{Which is half range sine series.}$$

$$\text{So, } c_n \left(\frac{n\pi c}{l}\right) = \frac{2}{l} \int_0^l g(x) \cdot \sin\left(\frac{n\pi x}{l}\right) dx$$

$$\Rightarrow c_n = \frac{2}{n\pi c} \int_0^l g(x) \cdot \sin\left(\frac{n\pi x}{l}\right) dx \quad ; n = 1, 2, 3, \dots$$

✓ Substituting b_n & c_n in eq. (8), we get required solution is,

$$\Rightarrow y(x, t) = \sum_{n=1}^{\infty} \left[b_n \cos\left(\frac{n\pi ct}{l}\right) + c_n \sin\left(\frac{n\pi ct}{l}\right) \right] \sin\left(\frac{n\pi x}{l}\right)$$

❖ SOLUTION OF HEAT EQUATION:

✓ The one – dimensional wave equation is

$$\frac{\partial y}{\partial t} = c^2 \frac{\partial^2 y}{\partial x^2} \quad \dots \dots (1)$$

✓ Boundary conditions:

$$y(0, t) = 0, \text{ for all } t. \quad \dots \dots (2)$$

$$y(l, t) = 0, \text{ for all } t. \quad \dots \dots (3)$$

✓ Initial Conditions

$$y(x, 0) = f(x) \quad \dots \dots (4)$$

✓ Suppose, $y(x, t) = X(x) \cdot T(t)$ is a solution of eq. (1).

$$\Rightarrow \frac{\partial y}{\partial t} = X T', \quad \frac{\partial^2 y}{\partial x^2} = X'' T$$

✓ Substituting in eq. (1)

$$\Rightarrow X T' = c^2 X'' T$$

$$\Rightarrow \frac{X''}{X} = \frac{1}{c^2} \frac{T'}{T} = k$$

✓ When, $k = 0$

$$\text{Now, } \frac{X''}{X} = k = 0$$

$$\Rightarrow X(x) = (c_1 + c_2 x)$$

$$\text{Now, } \frac{1}{c^2} \frac{T'}{T} = k = 0$$

$$\Rightarrow T(t) = c_3$$

Solution:

$$y(x, t) = X(x) \cdot T(t) = (c_1 + c_2 x) \cdot c_3$$

✓ When, k is positive. (i.e. $k = p^2$)

$$\text{Now, } \frac{X''}{X} = p^2$$

$$\Rightarrow D^2 X = X p^2$$

$$\Rightarrow (D^2 - p^2)X = 0$$

Auxiliary Equation,

$$m^2 - p^2 = 0$$

$$\Rightarrow m = +p, -p$$

$$\text{Now, } \frac{1}{c^2} \frac{T'}{T} = p^2$$

$$\Rightarrow \frac{T'}{T} = c^2 p^2$$

$$\Rightarrow \int \frac{T'}{T} dt = \int c^2 p^2 dt$$

$$\Rightarrow \log T = c^2 p^2 t + \log c_3$$

$$\text{So, } X(x) = c_1 e^{p x} + c_2 e^{-p x}$$

$$\Rightarrow \log\left(\frac{T}{c_3}\right) = c^2 p^2 t$$

$$\Rightarrow T = c_3 \cdot e^{c^2 p^2 t}$$

Solution:

$$y(x, t) = X(x) \cdot T(t) = (c_1 e^{p x} + c_2 e^{-p x})(c_3 \cdot e^{c^2 p^2 t})$$

✓ When, k is negative. (i.e. $k = -p^2$)

$$\text{Now, } \frac{X''}{X} = -p^2$$

$$\Rightarrow D^2 X = -X p^2$$

$$\Rightarrow (D^2 + p^2)X = 0$$

Auxiliary Equation,

$$m^2 + p^2 = 0$$

$$\Rightarrow m = +i p, -i p$$

$$\text{So, } X(x) = c_1 \cos p x + c_2 \sin p x$$

$$\text{Now, } \frac{1}{c^2} \frac{T'}{T} = -p^2$$

$$\Rightarrow \frac{T'}{T} = -c^2 p^2$$

$$\Rightarrow \int \frac{T'}{T} dt = - \int c^2 p^2 dt$$

$$\Rightarrow \log T = -c^2 p^2 t + \log c_3$$

$$\Rightarrow \log\left(\frac{T}{c_3}\right) = -c^2 p^2 t$$

$$\Rightarrow T = c_3 \cdot e^{-c^2 p^2 t}$$

Solution:

$$y(x, t) = X(x) \cdot T(t) = (c_1 \cos p x + c_2 \sin p x)(c_3 \cdot e^{-c^2 p^2 t})$$

✓ We are dealing with a problem on heat conduction, u must be a transient u is to decrease with the increase of time t . So, we can consider only 3rd case. i.e. $k < 0$.

$$y(x, t) = (c_1 \cos p x + c_2 \sin p x)(c_3 \cdot e^{-c^2 p^2 t}) \quad \dots \dots (6)$$

✓ We have, $y(0, t) = 0$

$$\Rightarrow c_1 (c_3 \cdot e^{-c^2 p^2 t}) = 0$$

$$\Rightarrow c_1 = 0$$

$$\text{So, we have } y(x, t) = c_2 \sin p x (c_3 \cdot e^{-c^2 p^2 t}) \quad \dots \dots (6)$$

✓ We have, $y(l, t) = 0$

$$\Rightarrow c_2 \sin p l (c_3 \cdot e^{-c^2 p^2 t}) = 0$$

$$\Rightarrow \sin pl = 0$$

$$\Rightarrow pl = n\pi$$

$$\Rightarrow p = \frac{n\pi}{l} ; n = 1, 2, 3, \dots$$

$$\text{Hence, } y(x, t) = c_2 \sin\left(\frac{n\pi x}{l}\right) \left(c_3 \cdot e^{-\frac{c^2 n^2 \pi^2 t}{l^2}}\right)$$

$$\Rightarrow y(x, t) = b_n \sin\left(\frac{n\pi x}{l}\right) e^{-\frac{c^2 n^2 \pi^2 t}{l^2}} \quad \text{Where, } b_n = c_2 c_3$$

✓ Taking sum of all solution for $n = 1, 2, 3, \dots$

$$\Rightarrow y(x, t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) e^{-\frac{c^2 n^2 \pi^2 t}{l^2}} \dots \dots (7)$$

From, eq. (4) and eq. (7),

$$\Rightarrow y(x, 0) = f(x) = \sum_{n=1}^{\infty} b_n \cdot \sin\left(\frac{n\pi x}{l}\right) \quad \text{Which is half range sine series.}$$

$$\text{So, } b_n = \frac{2}{l} \int_0^l f(x) \cdot \sin\left(\frac{n\pi x}{l}\right) dx \quad ; n = 1, 2, 3, \dots$$

✓ Substituting b_n in eq. (7), we get required solution is,

$$\Rightarrow y(x, t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) e^{-\frac{c^2 n^2 \pi^2 t}{l^2}}$$

✓ Because of the exponential factor all the terms in eq. (7) approach zero as t approaches to infinity. The rate of decay increases with n .

❖ SOLUTION OF LAPLACE EQUATION:

✓ The two – dimensional wave equation is

$$\frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \dots \dots (1)$$

✓ If the heat is steady means it is independent of time i.e. $\frac{\partial u}{\partial t} = 0$.

- ✓ Heat equation reduces to $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.
- ✓ Suppose, $y(x, t) = X(x) \cdot T(t)$ is a solution of eq. (1).

$$\Rightarrow \frac{\partial^2 y}{\partial x^2} = X''T, \frac{\partial^2 u}{\partial y^2} = X Y''.$$

- ✓ Substituting in eq. (1)

$$\Rightarrow X''Y + X Y'' = 0$$

$$\Rightarrow X''Y = -X Y''$$

$$\Rightarrow \frac{X''}{X} = -\frac{Y''}{Y} = k$$

- ✓ When, $k = 0$

$$X(x) = (c_1 + c_2 x) \text{ and } Y(y) = (c_3 + c_4 y)$$

Solution:

$$y(x, t) = X(x) \cdot Y(y) = (c_1 + c_2 x)(c_3 + c_4 y)$$

- ✓ When, k is positive. (i.e. $k = p^2$)

$$\text{Now, } \frac{X''}{X} = p^2$$

$$\Rightarrow D^2 X = X p^2$$

$$\Rightarrow (D^2 - p^2)X = 0$$

Auxiliary Equation,

$$m^2 - p^2 = 0 \Rightarrow m = +p, -p$$

$$\text{So, } X(x) = c_1 e^{p x} + c_2 e^{-p x}$$

$$\text{Now, } \frac{Y''}{Y} = -p^2$$

$$\Rightarrow D^2 Y = -Y p^2$$

$$\Rightarrow (D^2 + p^2)Y = 0$$

Auxiliary Equation,

$$m^2 + p^2 = 0 \Rightarrow m = i p, -i p$$

$$\text{So, } Y(y) = c_3 \cos py + c_4 \sin py$$

Solution:

$$u(x, y) = X(x) \cdot Y(y) = (c_1 e^{p x} + c_2 e^{-p x})(c_3 \cos py + c_4 \sin py)$$

- ✓ When, k is negative. (i.e. $k = -p^2$)

$$\text{Now, } \frac{X''}{X} = -p^2$$

$$\text{Now, } \frac{Y''}{Y} = p^2$$

$$\Rightarrow D^2X = -X p^2$$

$$\Rightarrow (D^2 + p^2)X = 0$$

Auxiliary Equation,

$$m^2 + p^2 = 0 \Rightarrow m = +i p, -i p$$

$$\text{So, } X(x) = c_1 \cos p x + c_2 \sin p x$$

$$\Rightarrow D^2Y = Y p^2$$

$$\Rightarrow (D^2 - p^2)Y = 0$$

Auxiliary Equation,

$$m^2 - p^2 = 0 \Rightarrow m = +p, -p$$

$$\text{So, } Y(y) = c_3 e^{py} + c_4 e^{-py}$$

Solution:

$$u(x, y) = X(x) \cdot Y(y) = (c_1 \cos px + c_2 \sin px)(c_3 e^{py} + c_4 e^{-py})$$

METHOD - 5: EXAMPLE ON WAVE, HEAT AND LAPLACE EQUATION

C	1	A string is stretched between two fixed points and the points of the string are given initial velocities $g(x)$, where $g(x) = \begin{cases} \frac{2x}{1}, & 0 < x < \frac{1}{2} \\ \frac{2(1-x)}{1}, & \frac{1}{2} < x < 1 \end{cases}$ x being the distance from the first end point. Find the displacement of the string at any time.	
H	2	A string of length $L = \pi$ has its ends fixed at $x = 0$ & $x = \pi$. At time $t = 0$, the string is given a shape defined by $f(x) = 50x(\pi - x)$, then it is released. Find the deflection of the string at any time t .	W 2019 (7)
C	3	Solve the equation $\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$ for the condition of heat along rod without radiation subject to the conditions (i) $\frac{\partial u}{\partial t} = 0$ for $x = 0$ and $x = 1$; (ii) $u = lx - x^2$ at $t = 0$ for all x .	W 2019 (7)
H	4	Find the solution of $u_t = c^2 u_{xx}$ together with the initial and boundary conditions $u(0, t) = u(2, t) = 0$; $t \geq 0$, $u(x, 0) = 10$; $0 \leq x \leq 2$.	S 2020 (7)
H	5	Find the temperature in the thin metal rod of length l with both the ends insulated and initial temperature is $\sin \frac{\pi x}{l}$.	W 2019 (4)

C	6	Solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ which satisfies the conditions $u(0, y) = u(l, y) = u(x, 0) = 0$ and $u(x, b) = \sin \frac{n\pi x}{l}$.	
H	7	Derive the one dimensional wave equation that governs small vibration of an elastic string. Also state physical assumptions that you make for the system.	W 2019 (7)
C	8	Find the solution of wave equation $u_{tt} = c^2 u_{xx}$, $0 \leq x \leq \pi$ with initial and boundary conditions $u(0, t) = u(\pi, t) = 0$; $t > 0$, $u(x, 0) = k(\sin x - \sin 2x)$, $u_t(x, 0) = 0$; $0 \leq x \leq \pi$, ($c^2 = 1$).	S 2020 (7)



GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– III (New) EXAMINATION – WINTER 2019****Subject Code: 3130005****Date: 26/11/2019****Subject Name: Complex Variables and Partial Differential Equations****Time: 02:30 PM TO 05:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		Marks
Q.1	(a) Find the real and imaginary parts of $f(z) = \frac{3i}{2+3i}$.	03
	(b) State De-Moivre's formula and hence evaluate $(1+i\sqrt{3})^{100} + (1-i\sqrt{3})^{100}$.	04
	(c) Define harmonic function. Show that $u(x, y) = \sinh x \sin y$ is harmonic function, find its harmonic conjugate $v(x, y)$.	07
Q.2	(a) Determine the Mobius transformation which maps $z_1 = 0, z_2 = 1, z_3 = \infty$ into $w_1 = -1, w_2 = -i, w_3 = 1$.	03
	(b) Define $\log z$, prove that $i^i = e^{-(4n+1)\frac{\pi}{2}}$.	04
	(c) Expand $f(z) = \frac{1}{(z-1)(z+2)}$ valid for the region (i) $ z < 1$ (ii) $1 < z < 2$ (iii) $ z > 2$.	07
	OR	
	(c) Find the image of the infinite strips (i) $\frac{1}{4} \leq y \leq \frac{1}{2}$ (ii) $0 < y < \frac{1}{2}$ under the transformation $w = \frac{1}{z}$. Show the region graphically.	07
Q.3	(a) Evaluate $\int_c (x - y + ix^2) dz$ where c is a straight line from $z = 0$ to $z = 1 + i$.	03
	(b) Check whether the following functions are analytic or not at any point, (i) $f(z) = 3x + y + i(3y - x)$ (ii) $f(z) = z^{3/2}$.	04
	(c) Using residue theorem, evaluate $\int_0^\infty \frac{dx}{(x^2+1)^2}$.	07
	OR	
Q.3	(a) Expand Laurent series of $f(z) = \frac{1-e^z}{z}$ at $z = 0$ and identify the singularity.	03
	(b) If $f(z) = u + iv$, is an analytic function, prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) Re f(z) ^2 = 2 f'(z) ^2$.	04
	(c) Evaluate the following: i. $\int_c \frac{z+3}{z-1} dz$ where c is the circle (a) $ z = 2$ (b) $ z = \frac{1}{2}$. ii. $\int_c \frac{\sin z}{\left(z-\frac{\pi}{4}\right)^3} dz$ where c is the circle $ z = 1$.	07

- Q.4** (a) Evaluate $\int_0^{2+4i} \operatorname{Re}(z) dz$ along the curve $z(t) = t + it^2$. **03**
- (b) Solve $x^2 p + y^2 q = (x + y)z$. **04**
- (c) Solve the equation $\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$ for the condition of heat along rod without radiation subject to the conditions (i) $\frac{\partial u}{\partial t} = 0$ for $x = 0$ and $x = l$; **07**
(ii) $u = lx - x^2$ at $t = 0$ for all x .
- OR**
- Q.4** (a) Solve $\frac{\partial^2 z}{\partial x^2} + 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = e^{2x+3y}$. **03**
- (b) Solve $px + qy = pq$ using Charpit's method. **04**
- (c) Find the general solution of partial differential equation $u_{xx} = 9u_y$ using method of separation of variables. **07**
- Q.5** (a) Using method of separation of variables, solve $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$. **03**
- (b) Solve $z(xp - yq) = y^2 - x^2$. **04**
- (c) A string of length $L = \pi$ has its ends fixed at $x = 0$ and $x = \pi$. At time $t = 0$, the string is given a shape defined by $f(x) = 50x(\pi - x)$, then it is released. Find the deflection of the string at any time t . **07**
- OR**
- Q.5** (a) Solve $p^3 + q^3 = x + y$. **03**
- (b) Find the temperature in the thin metal rod of length l with both the ends insulated and initial temperature is $\sin^{\pi x}/l$. **04**
- (c) Derive the one dimensional wave equation that governs small vibration of an elastic string . Also state physical assumptions that you make for the system. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– III EXAMINATION – SUMMER 2020****Subject Code: 3130005****Date: 27/10/2020****Subject Name: Complex Variables and Partial Differential Equations****Time: 02:30 PM TO 05:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		Marks
Q.1	(a) If $u = x^3 - 3xy$ is find the corresponding analytic function $f(z) = u + iv$.	03
	(b) Find the roots of the equation $z^2 - (5 + i)z + 8 + i = 0$.	04
	(c) (i) Determine and sketch the image of $ z = 1$ under the transformation $w = z + i$.	03
	(ii) Find the real and imaginary parts of $f(z) = z^2 + 3z$.	04
Q.2	(a) Evaluate $\int_C (x^2 - iy^2) dz$ along the parabola $y = 2x^2$ from (1,2) to (2,8).	03
	(b) Find the bilinear transformation that maps the points $z = \infty, i, 0$ into $w = 0, i, \infty$.	04
	(c) (i) Evaluate $\oint_C \frac{e^{-z} dz}{z+1}$, where C is the circle $ z = 1/2$.	03
	(ii) Find the radius of convergence of $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{n^2} z^n$.	04
	OR	
	(c) (i) Find the fourth roots of -1 .	03
	(ii) Find the roots of $\log z = i \frac{\pi}{2}$.	04
Q.3	(a) Find $\oint_C \frac{1}{z^2} dz$, where $C : z = 1$.	03
	(b) For $f(z) = \frac{1}{(z-1)^2(z-3)}$, find Residue of $f(z)$ at $z=1$.	04
	(c) Expand $f(z) = \frac{1}{(z+2)(z+4)}$ in a Laurent series for the regions (i) $ z < 2$, (ii) $2 < z < 4$, (iii) $ z > 4$.	07
	OR	
Q.3	(a) Find $\oint_C \frac{z+4}{z^2+2z+5} dz$, where $C : z+1 = 1$.	03
	(b) Evaluate using Cauchy residue theorem $\int_C \frac{e^{2z}}{(z+1)^3} dz$; $C: 4x^2 + 9y^2 = 16$.	04
	(c) Expand $f(z) = \frac{1}{z(z-1)(z-2)}$ in Laurent's series for the regions (i) $ z < 1$, (ii) $1 < z < 2$, (iii) $ z > 2$.	07

- Q.4** (a) Solve $xp + yq = x - y$. **03**
 (b) Derive partial differential equation by eliminating the arbitrary constants a and b from $z = ax + by + ab$. **04**
 (c) (i) Solve the p.d.e. $2r + 5s + 2t = 0$. **03**
 (ii) Find the complete integral of $p^2 = qz$. **04**
OR
- Q.4** (a) Find the solution of $x^2p + y^2q = z^2$. **03**
 (b) Form the partial differential equation by eliminating the arbitrary function ϕ from $z = \phi\left(\frac{y}{x}\right)$. **04**
 (c) (i) Solve the p.d.e. $(D^2 - D'^2 + D - D')z = 0$. **03**
 (ii) Solve by Charpit's method $yzp^2 - q = 0$. **04**
- Q.5** (a) Solve $(2D^2 - 5DD' + 2D'^2)z = 24(y - x)$. **03**
 (b) Solve the p.d.e. $u_x + u_y = 2(x + y)u$ using the method of separation of variables. **04**
 (c) Find the solution of the wave equation $u_{tt} = c^2u_{xx}$, $0 \leq x \leq \pi$ with the initial and boundary conditions $u(0, t) = u(\pi, t) = 0; t > 0$, $u(x, 0) = k(\sin x - \sin 2x), u_t(x, 0) = 0; 0 \leq x \leq \pi$. ($c^2 = 1$) **07**
OR
- Q.5** (a) Solve the p.d.e. $r + s + q - z = 0$. **03**
 (b) Solve $2u_x = u_t + u$ given $u(x, 0) = 4e^{-3x}$ using the method of separation of variables. **04**
 (c) Find the solution of $u_t = c^2u_{xx}$ together with the initial and boundary conditions $u(0, t) = u(2, t) = 0; t \geq 0$ and $u(x, 0) = 10; 0 \leq x \leq 2$. **07**



GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering

Subject Code: 3130005

Semester – III

Subject Name: Complex Variables and Partial Differential Equations

Type of course: Basic Science Course

Prerequisite: Geometry, trigonometry, calculus and ODE.

Rationale: This subject is a powerful tool for solving a wide array of applied problems.

Teaching and Examination Scheme:

Teaching Scheme			Credits	Examination Marks				Total Marks	
L	T	P		C	Theory Marks		Practical Marks		
					ESE (E)	PA (M)	ESE (V)		PA (I)
3	2	0	5	70	30	0	0	100	

Content:

Sr. No.	Content	Total Hrs	% Weightage
01	Polar Form of Complex Numbers, Powers and Roots, Complex Variable – Differentiation : Differentiation, Cauchy-Riemann equations, analytic functions, harmonic functions, finding harmonic conjugate; elementary analytic functions (exponential, trigonometric, logarithm) and their properties; Conformal mappings, Mobius transformations and their properties.	12	28%
02	Complex Variable - Integration : Contour integrals, Cauchy-Goursat theorem (without proof), Cauchy Integral formula (without proof), Liouville's theorem and Maximum-Modulus theorem (without proof); Sequences, Series, Convergence Tests, Power Series, Functions Given by Power Series, Taylor and Maclaurin Series, Uniform Convergence.	08	20%
03	Laurent's series; Zeros of analytic functions, singularities, Residues, Cauchy Residue theorem (without proof), Residue Integration Method, Residue Integration of Real Integrals.	06	14%
04	First order partial differential equations, solutions of first order linear and nonlinear PDEs, Charpit's Method	06	14%
05	Solution to homogeneous and nonhomogeneous linear partial differential equations second and higher order by complementary function and particular integral method. Separation of variables method to simple problems in Cartesian coordinates, second-order linear equations and their classification, Initial and boundary conditions, Modeling and solution of the Heat, Wave and Laplace equations.	10	24 %

Suggested Specification table with Marks (Theory):

Distribution of Theory Marks					
R Level	U Level	A Level	N Level	E Level	C Level
7	28	35	0	0	0

Legends: R: Remembrance; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create and above Levels (Revised Bloom's Taxonomy)



GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering

Subject Code: 3130005

Note: This specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from above table. This subject will be taught by Maths faculties.

Reference Books:

- (1) Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, John Wiley and Sons.
- (2) Peter O'Neill, Advanced Engineering Mathematics, 7th Edition, Cengage.
- (3) Dennis G. Zill, 4th edition, Advanced Engineering Mathematics, 4th Edition, Jones and Bartlett Publishers.
- (4) Dennis G. Zill, Patrick D. Shanahan, A First Course in Complex Analysis with Applications, Jones and Bartlett Publishers.
- (5) S. J. Farlow, Partial Differential Equations for Scientists and Engineers, Dover Publications, 1993.
- (6) Ian Sneddon, Elements of Partial Differential Equations, McGraw Hill.
- (7) J. W. Brown and R. V. Churchill, Complex Variables and Applications, McGraw Hill.

Course Outcome:

Sr. No.	CO statement	Marks % weightage
CO-1	convert complex number in a polar form, plot the roots of a complex number in complex plane, find harmonic conjugate of analytic functions and apply conformal mapping in geometrical transformation	28%
CO-2	evaluate complex integration by using various result, test convergence of complex sequence and series and expand some analytic function in Taylor's series	20%
CO-3	find Laurent's series and pole of order, and apply Cauchy Residue theorem in evaluating some real integrals	14%
CO-4	form and solve first order linear and nonlinear partial differential equations	14%
CO-5	apply the various methods to solve higher order partial differential equations, modeling and solve some engineering problems related to Heat flows, Wave equation and Laplace equation	24 %

List of Open Source Software/learning website:

MIT Opencourseware. NPTEL.